semanticfa: An R Package for Response-Free Semantic Factor Analysis of Psychometric Scales

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Abstract

The items of a psychometric scale constitute a proposal, written in ordinary language, about the structure of a construct—and language-model embeddings make that proposal measurable before a single respondent is recruited. We introduce `semanticfa`, an R package for semantic factor analysis (SFA): exploratory factor analysis of the similarity structure of language-model embeddings of scale items, requiring no response data at any stage. The package implements the complete workflow: similarity construction under keyed, keying-free, and natural-language-inference encodings; factor retention by multi-criterion consensus; extraction with Monte Carlo–calibrated diagnostics; interpretation by anchoring, semantic projection, congruence, and jingle–jangle screening; and response-free refinement (redundancy detection, short forms, candidate-item vetting). A fully reproducible demonstration exercises every exported function on the 50-item International Personality Item Pool Big-Five Factor Markers, embedding the items with Qwen3-Embedding-8B and validating the semantic structure against a conventional factor analysis of 874,434 respondents. The retention consensus indicated five factors, and the semantic solution recovered the Big Five domains, with congruence to the human factors graded by domain (Tucker's ϕ from .92 for Openness to .52 for Agreeableness) and semantic similarities correlating $r = .46$ with response correlations across all 1,225 item pairs. Sign-flipping reverse-keyed embeddings produced anti-topic vectors that aligned worst with keyed human data, and a semantically central 25-item short form recovered the theoretical structure better than the full inventory. We frame the agreement of the two structures—semantic–behavioral coherence—as an estimable relation in a construct's nomological network: a complement to, not a replacement for, response-based factor analysis. The package is available on CRAN.

Keywords: semantic factor analysis, language models, embeddings, construct validity, scale development, R package

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A psychometric scale is two things at once: to the respondent, prompts to be answered; to the discipline, a theory in miniature, each item a public, ordinary-language commitment about what the target construct involves. Classical factor analysis examines the first—the covariance structure of responses—and can begin only after people answer. This article concerns the second: the structure of the items' meaning, which predates any respondent and which language-model embeddings now make measurable. We introduce `semanticfa`, an R package for semantic factor analysis (SFA)—exploratory factor analysis of the similarity structure of such embeddings, with no response data at any stage—and demonstrate the complete workflow on the 50-item International Personality Item Pool (IPIP) Big-Five Factor Markers, validating the semantic structure against a conventional factor analysis of more than 870,000 respondents. The Introduction gives a theoretical account of analyzing a scale's language, proposing semantic–behavioral coherence—agreement between the two structures—as a relation in the construct's nomological network; reviews the consolidating literature on language models and psychometrics by what each line of work does; and introduces the package and the demonstration.

Constructs and the Nomological Network

Psychological constructs cannot be inspected directly; they are postulated attributes inferred from regularities in what people do, feel, and report. Cronbach and Meehl (1955) formalized this as construct validity: where no criterion or content universe fully defines an attribute, a test is validated only by the construct's place in a nomological network of lawful relations to indicators, experimental manipulations, and other constructs. Stating a construct's laws is saying what it is, elaborating them is learning more, and a sparse network leaves a vague construct: meaning is relational—the evidential web's breadth and coherence, not a hidden essence. Messick (1995) recast all validation as construct validation—an integrated evaluative judgment about score interpretations, not a property of instruments—with six aspects of evidence (content, substantive, structural, generalizability, external, consequential) and two pervasive failure modes: construct underrepresentation and construct-irrelevant variance. Clark and Watson (2019) make the network a working tool: a precise written conceptualization, boundaries with near neighbors included, is scale development's first and most consequential step, for without an articulated theory there is no construct validity to investigate. Borsboom et al. (2004) dissent on ontological grounds: a test is valid only if its attribute exists and causally produces variation in test outcomes, and psychology's networks are too loosely specified to fix the meanings of theoretical terms. We adopt the relational framework but concede the realist point: no correlational pattern, behavioral or semantic, by itself certifies that an attribute exists; both structures figure here as evidence about how a construct is defined and how its indicators behave, not as proof of its reality.

Two Relational Structures Over the Same Items

Psychology's statistical machinery for such relational webs predates construct-validity vocabulary by half a century. Holding character "a definite and durable 'something' " and so measurable, Galton (1884, p. 181) mapped it from the dictionary, not behavior, estimating in Roget's Thesaurus "fully one thousand words expressive of character," distinct yet overlapping in meaning—founding the lexical tradition and anticipating the semantic-similarity analyses below. Spearman (1904) supplied the inferential engine of "correlational psychology": diverse intellectual performances correlate positively; their common variance marks a single general function, estimable with its per-measure saturation from the intercorrelation matrix. Factor analysis—low-dimensional structure from a high-dimensional relational matrix—descends from this move; Jöreskog (1969) made it confirmatory—any subset of loadings, factor correlations, and uniquenesses fixed in advance, the rest maximum-likelihood estimated with a likelihood-ratio test of fit—and measurement models testable hypotheses. Goldberg (1990) rejoined the threads: virtually identical five-factor structures emerged across 1,431 trait adjectives, ten extraction–rotation combinations, and self- and peer ratings, establishing the Big Five as a robust summary of the English trait lexicon. The object was always a relational matrix whose variables, in the lexical tradition, were language; semantic factor analysis changes the source, not the logic.

A construct enters empirical practice through its items—the researcher's most concrete public commitment, in ordinary language, to the experiences, behaviors, and situations it involves: for Messick (1995), the content aspect of validity. Reflective or formative (Bollen & Lennox, 1991), items encode the intended meaning more exactly than any definition. If, as Wittgenstein (1953, §43) held for a large class of cases, "the meaning of a word is its use in the language," that meaning is public and analyzable—regularities of use across the language community, prior to and independent of any respondent sample; and a concept needs no sharp boundaries or common essence to do scientific work, its instances sharing only family resemblances (Wittgenstein, 1953, §§66–67)—apt for constructs whose items cohere without one shared feature. Two relational structures over the same items follow: items cluster behaviorally in the person-by-item response matrix because the same people endorse them, and semantically in the language itself because their words are used alike—recovered, respectively, by classical and semantic factor analysis. Nor is wording behaviorally peripheral: Clark and Watson (2019) observe that exact phrasing can greatly influence the construct measured—almost any negative mood term virtually guarantees a substantial negative-affectivity component—and that a single item carries interpretable variance at several construct-hierarchy levels; a method analyzing wording directly is thus a missing instrument.

Semantic–Behavioral Coherence as a Nomological Relation

Because both structures are factor solutions over the same items, their comparison has a nomological home: Cronbach and Meehl (1955) held that qualitatively different operations measure the same thing when the network ties them to the same construct, and agreement between meaning-based and response-based solutions is just such convergence—disjoint data sources, one needing no respondents, one partition of the items. We call such agreement *semantic–behavioral coherence*, an estimable scale property, not a validity requirement: divergence is expected, and diagnostic, under at least three conditions. First, formative indicator sets: Bollen and Lennox (1991) show that the measurement model leaves causal-indicator correlations unconstrained, so semantically related items may barely cluster and an item census, not a sample, is required. Second, causal-system constructs: Borsboom (2008) argues that some clinical constructs are networks of mutually causal symptoms—panic attacks breed worry, worry avoidance—that constitute the disorder rather than measure a common cause, so behavioral covariance tracks the causal chain and can outrun semantic kinship. Third, construct-irrelevant variance: method factors and wording artifacts add behavioral covariance with no semantic counterpart (Messick, 1995). There is quantitative precedent: in 39,775 responses to the 42-item Depression Anxiety Stress Scales, Stanghellini et al. (2024) found non-random alignment between item-text semantic communities (syntactic and synonym networks) and exploratory-graph-analysis factors, quantified as “semantic loadings”—Jaccard overlaps—and proposed as a lens on content and discriminant validity. We draw the same comparison with embedding-based semantic structure and explicit congruence statistics, promoting it from diagnostic device to named nomological relation.

Measuring Meaning: From Word Co-Occurrence to Sentence Embeddings

Semantic factor analysis stands or falls with a well-tested claim: language-model embeddings quantify what items mean. Distributional semantic models operationalize the meaning-as-use thesis introduced above; psychology built latent semantic analysis and the hyperspace analogue to language as implemented cognitive theories of semantic memory, not engineering conveniences (Günther et al., 2019). `word2vec`'s shallow word-from-context prediction over billions of tokens (Mikolov et al., 2013) yields spaces where proximity tracks similarity of use and arithmetic recovers relational structure—a geometry Pennington et al. (2014) matched by weighted co-occurrence factorization. The geometry behaves like a psychological quantity: error-driven, associatively trained embeddings explain semantic priming (matching or exceeding association- and feature-norm predictors), similarity judgment, and word association (Mandera et al., 2017); co-occurrence distances predicted lexical decision and naming latencies before neural embeddings (Buchanan et al., 2001); and embeddings encode affective, real-world spatial, and social structure (see Günther et al., 2019).

Items, however, are sentences. Transformer self-attention contextualizes every token (Vaswani et al., 2017), and the masked-language pre-training of bidirectional transformers by Devlin et al. (2019) yields representations transferable across language-understanding tasks; yet pooled BERT outputs track human sentence-similarity judgments worse than averaged GloVe vectors, so Sentence-BERT fine-tunes a siamese transformer on natural-language-inference (NLI) corpora, mapping sentences to fixed-length, cosine-comparable vectors (Reimers & Gurevych, 2019). Such embeddings are the method's raw material—each item a point in a common space, the pairwise-similarity matrix a summary of the scale's semantic organization—and because cosine similarity is symmetric while entailment and contradiction are not, `semanticfa` treats inference-based relations as a separate measurement channel.

The geometry licenses factor analysis only if its latent structure matches what humans recognize and report; two findings say it does. Westbury (2025), treating embeddings as compressed descriptions of words' contexts of use, not arbitrary symbols, reports that large language models reproduce human meaningfulness ratings of novel sentences at $r = .66$—roughly half the reliable variance, hard to square with mere recombination of surface statistics. Most directly, Kmetty et al. (2021) factor analyzed occupation-title embeddings' cosine-similarity matrix instead of the correlation matrix; the leading dimension correlated approximately .7 with the survey-based International Socio-Economic Index of occupational status and replicated at $r = .97$ across independently trained embedding spaces. Occupations are not personality items, but extracting latent dimensions from embedding geometry and validating them against respondent-derived structure is precisely what `semanticfa` systematizes.

Predicting Response Structure From Item Language

Empirical inter-item correlations can be predicted from item text alone. Hommel and Arslan (2025) fine-tuned a sentence transformer to make contradictions negatively similar, adapted it to roughly 90,000 public-survey item pairs, and preregistered predictions for 246 novel personality, clinical, social, occupational, and attitude items; disattenuated accuracy was .71 (pilot holdout) and .59 (validation) for item-pair correlations, .84 for scale intercorrelations and reliabilities alike—pilot-study error margins at zero cost and delay. Even general-purpose embeddings carry much of the signal: OpenAI text-embedding cosines correlated .77 with the 42-item Depression Anxiety Stress Scales' inter-item correlations and put the empirically most correlated partner in an item's three nearest neighbors for 95% of items (Ravenda et al., 2025); a personality-tuned MPNet encoder reached .67 on the 96 VIA Inventory of Strengths items (31,697 respondents) (Feraco & Toffalini, 2025). The relation is uneven: multilevel analysis of 463 item pairs in nine construct dimensions found prediction on average but substantial variation, weakest where baseline cohesion was strongest (Zhang & Qiu, 2026).

If embedding similarities approximate correlations, the factor-analytic toolbox applies before any respondent is recruited: principal components of a RoBERTa cosine matrix grouped the 50-item Big Five markers questionnaire analyzed here by intended construct, with embedding–response loading correlations above .40 (Spearman) within constructs (Casella et al., 2024). In four inventories (Big Five markers, RIASEC, Humor Styles Questionnaire, DASS-42), Milano, Luongo, et al. (2025) found over 70% of items on average loading highest on the assigned construct's component—within-construct correspondence moderate for the Big Five (.40–.65), strong for the DASS (.80–.90)—and termed this "ex ante" validation. Formalizing the premise—embeddings substituting for response vectors—Guenole et al. (2025) found exploratory "pseudo-factor analysis" of facet-level cosine matrices recovering IPIP-NEO-300 and HEXACO-240 configural structure, though Tucker congruence with empirical loadings generally fell short of equivalence even where factors correlated strongly. Confirmatory variants are conservative: every scale model that fit the semantic matrix in Feraco and Toffalini (2025) also fit the empirical one, but roughly half of semantic misfit verdicts were false alarms—still a useful pre-collection screen.

A recurring obstacle is reverse keying: pretrained embeddings almost never assign negative similarities—before fine-tuning, Hommel and Arslan (2025) found "I am the life of the party" no less similar to its polar opposite than to a paraphrase. Some rescore reverse-keyed human responses before comparing structures (Casella et al., 2024; Milano, Luongo, et al., 2025); others adjust the semantic side: contradiction training in Hommel and Arslan (2025) improved negatively worded items but left a positive-sign bias (67% of synthetic correlations positive vs. 59% empirical), and sign-flipping embeddings by theoretical key in Guenole et al. (2025) improved configural recovery for the NEO but fared worst for the HEXACO, consistently blending Emotionality into Honesty–Humility. These partly contradictory remedies make keying a substantive modeling choice, not a nuisance parameter; the demonstration below tests keying-free, sign-flipped, and entailment-based encodings against the same human benchmark.

Machine judgments also rival the experts who would otherwise supply them: Schoenegger et al. (2025) had 254 laypeople, 272 psychology experts, GPT-4o, Claude 3

Opus, and a deep network trained on empirical personality correlations predict 249 SAPA item-pair correlations; every artificial system beat most individual humans—the network topped over 99% of experts head to head—though median-aggregated expert predictions were statistically indistinguishable from it, and the two general-purpose models trailed both. Humans keep an edge in content validation: embedding models beat a graduate-level panel at assigning items to constructs on the lexically phrased Big Five Inventory (up to 97.5% for a personality-tuned encoder vs. 72%) but not on the behaviorally phrased Big Five Questionnaire (best model 80% vs. 82%) (Milano, Ponticorvo, & Marocco, 2025)—how surface language carries content conditions what embeddings recover. Throughout, what is embedded is the scale's language, not the people who answer it; a parallel literature predicts Big Five traits from roughly a million Reddit posts (Maharjan et al., 2025).

Taxonomy, Jingle–Jangle Fallacies, and Construct Redundancy

Embeddings also probe the field's taxonomy: scales with different content share a label (the jingle fallacy); near-interchangeable ones bear different labels (the jangle fallacy). Wulff and Mata (2025) co-embedded 4,452 IPIP items, 459 scales, and 277 construct labels with a fine-tuned sentence transformer whose similarities predicted empirical internal consistencies ($r = .75$ in sample, .61 out of sample), recovered most inventories' factor structures, tracked item and scale intercorrelations, and made misalignment measurable—an automated audit flagged 340 jingle and 164 jangle candidates among IPIP scale pairs, a count that clustering and relabeling more than halved with roughly a quarter as many labels. Wulff and Mata (2026) add three applications—mapping measure relations, relabeling taxonomies, and generating new items or constructs vetted in the same space—cautioning that semantic proximity does not settle theoretical identity (subjective and physiological stress may remain conceptually distinct) and that automated consolidation should inform, not replace, expert judgment.

The same geometry detects construct redundancy: Huang et al. (2025) embedded items with a large proprietary model, treated cosine similarities as "synthetic correlations," and applied agglomerative clustering, collecting no responses. Against theory-based benchmarks it separated unrelated constructs—conscientiousness from gratitude, within-cluster similarities far exceeding between-cluster ones—and reproduced a documented redundancy: grit's perseverance-of-effort facet joined conscientiousness, not its own scale, where meta-analytic response data locate the overlap. Output was stable across input configurations and closer to the expected structure than prompted chatbots—reproducible semantic analysis requires an explicit embedding-plus-clustering pipeline, not ad hoc prompting.

Clinical Structure From Embeddings

Clinical assessment is these methods' sternest test; two recent studies mark its promise and boundary. Kambeitz et al. (2025) embedded the items of four established psychopathology questionnaires—depression, anxiety, and stress symptoms, schizotypy, autism-related traits, and attenuated psychosis—and compared item-pair embedding similarities with empirical correlations in four datasets (ns of 1,099 to 39,755): the two correlated in every questionnaire ($r = .18$ to $.57$), machine-learning prediction raised this to moderate-to-high, and embedding clusters matched (by adjusted Rand index) both empirical clusters and published subdomains, semantic consistently beating sentiment-based embeddings. Kojima et al. (2026) pressed the most consequential case—whether the general psychopathology factor (p-factor) belongs to syndromes or to assessment language—by factor-analyzing 145 symptom items from eight instruments under item-embedding cosines and polychoric correlations from 985 community respondents (matrix correlation $r = .55$): both yielded a dominant general factor with nearly interchangeable loadings (congruence = .97), yet no specific factor reached the .90 benchmark. The divergences were informative: responses fused insomnia with somatic complaints—attributed to shared physiology, not shared meaning—while embeddings grouped items merely sharing wording; response-based factor scores predicted lifetime diagnoses better than text-based ones; weighting analyses found the textual information effectively a subset of the response-based. The lesson calibrates the enterprise: instrument language genuinely shadows empirical structure, but lived co-occurrence holds clinically informative variance beyond what wording predicts.

Methods and Pipelines for Embedding-Based Psychometrics

Shared linguistic features inflate raw cosines, masking the negative associations of reverse keying and opposed constructs; SQuID questionnaire-mean centering (Pellert et al., 2026)—`semanticfa`'s `squid` encoding—restores them without fine-tuning or re-annotation. So centered, a general-purpose model widened inter-item correlation ranges on three personality inventories and rivaled human raters on the Revised Portrait Value Questionnaire circumplex: 55% of dimension–dimension similarity variance explained, multidimensional scaling matching the pooled 49-country solution (Procrustes congruences .88, .82).

Pokropek (2026) formalized the confirmatory step: cosines of mean-centered vectors being Pearson correlations, centered similarity matrices enter structural-equation software with loadings, fit indices, reliabilities, and multigroup invariance tests. Latent means from eleven weekly war-related Twitter corpora tracked independently surveyed war anxiety over six overlapping time points (r = .873); yet 47.9% of four-indicator one-factor models of random words from the same thematically constrained corpus cleared the .95 CFI cutoff and 95% the .08 SRMR cutoff, so fit must be judged against chance—hence `semanticfa`'s Monte Carlo calibration.

In LLM-generated personality pools, the eigenvalue-greater-than-one rule and even principal-component parallel analysis overestimated dimensionality (mean bias errors of 15.06 and 7.37 dimensions, never correct), whereas Exploratory Graph Analysis (EGA) recovered generating structure at .95 normalized mutual information (NMI), 95.6% correct after filtering redundant and unstable items (Garrido et al., 2025); Golino (2026) added that item-embedding partitions peak in accuracy (NMI) and organization (Total Entropy Fit Index; TEFI) at opposite (shallow versus deep) coordinate depths, so optimizing either alone sacrifices the other—grounds for `sfa_nfactors()`'s four-criterion consensus.

Near-paraphrase items violate local independence and distort structure, acutely in large or machine-generated pools. Unique Variable Analysis (UVA; Christensen et al., 2023), flagging such pairs on a partial-correlation network, best balanced detection and false discovery across simulated continuous, polytomous, and dichotomous data (critical success index .880, false discovery rate .020), its fixed cutoff beating significance-based rules; built for response data yet adopted in embedding pipelines (Garrido et al., 2025; Russell-Lasalandra et al., 2026), it is the template for `sfa_redundancy()`.

Grand et al. (2022) showed embedding geometry supports direct, scale-like measurement: projecting word vectors orthogonally onto the antonym-anchor line (*small* to *big*) recovers graded human feature judgments (median correlation .47 across 52 category–feature pairs, significant in 32), whereas single-anchor projection collapses to a median of .06—the difference vector, not either endpoint, is the diagnostic direction; `sfa_project()` carries that projection to scale items and factors on researcher-defined pole pairs, probing factor meaning beyond loadings.

Scale abbreviation, among the most practical payoffs, conventionally presupposes an administered instrument. Clustering embeddings of the 50 IPIP Big-Five Factor Marker items analyzed here, Jung and Seo (2025) derived a response-free 30-item short form matching the original factor assignment at 96% and full-scale scores at $r = .95$–$.98$, rivaling classical-test-theory and genetic-algorithm short forms. Wang et al. (2026) added density-based clustering that discovers rather than fixes the factor count, topic-model keywords keeping clusters inspectable: across the Depression Anxiety Stress Scales, the IPIP pool, and a Chinese-language adolescent well-being measure, item counts fell 60.5% on average while confirmatory structure and inter-factor correlations held (adjusted Rand cluster–factor agreement .745–1.00). Both first rewrote reverse-keyed items as positive paraphrases—an ad hoc remedy for the keying problem examined head-on here—whereas `sfa_simplify()` builds short forms on keying-free encodings that need no paraphrase.

Generative Language Models as Respondents, Raters, and Test Developers

Others prompt generative models to act as respondents. Liu et al. (2025) calibrated six models' college-algebra answers with item response theory: the best model's difficulty parameters tracked human calibrations (Spearman $\rho = .87$, GPT-3.5); mixing 50 humans with as many resampled synthetic respondents raised difficulty-ordering recovery from $\rho = .89$ to .93 at higher root-mean-square error; and far narrower ability distributions (standard deviations 0.29–0.58 versus the human 0.98) rule out any single model as a current substitute. The "virtual respondents" of Lim et al. (2025)—mediating characteristics letting one trait yield different item answers—screened machine-generated Big Five, Schwartz values, and character-strength items into sets in the top 1% (Big Five) to 13% (Schwartz and VIA) of possible selections on human-derived convergent-validity criteria without claiming to replicate human psychology; pre-data vetting is feasible but only as faithful as its simulation.

Educational measurement instead predicts item parameters from item text: a 37-article (46-study) systematic review of large-scale assessment traces difficulty modeling from hand-crafted linguistic features to transformers, the strongest correlating .87 with empirical difficulty at root-mean-square errors as low as 0.165 (Peters et al., 2025). The signal is uneven: fine-tuned transformers predicted 1,119 operational English Language Arts items' banked difficulty at correlations near .72, yet on discrimination a gradient-boosted tree over structured item attributes ($r = .54$) beat every fine-tuned model (Han et al., 2025); a generative model fine-tuned to reproduce response probabilities at twenty discrete ability levels, however, recovers both parameters through synthetic item characteristic curves, discrimination especially well (Ormerod, 2026).

Generative models have also been validated as raters of constructs in text: in two stages—iterative prompt development, then confirmatory testing on manually coded held-out data—Bunt et al. (2025) found GPT-4o excellent on directly observable phenomena (reported speech and conversational repair, $F_1 = .91$ for the best prompts on both) but only moderate on an inferential ordinal construct, harm in healthcare complaints (weighted $\kappa = .49$–$.57$); valid machine coding requires actively establishing semantic, predictive, and content validity. Eberhardt et al. (2025) applied classical scale construction—theory-driven item pool, automated selection, confirmatory factor analysis, reliability estimation—to ratings from a locally hosted model rather than human judges: scored over 1,131 automatically transcribed therapy sessions, their eight-item "LLM rating scale" for patient engagement showed excellent internal consistency ($\omega = .95$) and theoretically coherent correlations with patient motivation ($r = .41$), therapist-rated between-session effort ($r = .39$), and later symptom outcome ($r = -.30$)—direct precedent that psychometric machinery transfers to language-model output.

A practical guide spans whole-instrument construction: item development (training-data quality, multi-model comparison, output validation) and "data-less" calibration, where sentence encoders check construct boundaries and item–definition alignment, embedding-similarity matrices are factor-analyzed, and, absent any sample, fit rests on model-free residual diagnostics (root-mean-square residual, common part accounted for) (Suárez-Álvarez et al., 2026); yet it cautions that embedding-based loadings—mostly positive cosine similarities—do not yet differentiate reverse-keyed items as conventional loadings do, leaving the human-response factor structure the gold standard for semantic structures. With a structured zero-shot prompt, Grobelny et al. (2025) had GPT-4 adapt an organizational-commitment scale from English to Polish; among 785 working adults randomized to either version, the adaptation equaled or surpassed the established human one in factor structure, measurement invariance, and convergent and discriminant validity, its internal consistency statistically lower ($\alpha = .89$ versus $.92$) yet within accepted standards.

These routes—simulated respondents (Lim et al., 2025; Liu et al., 2025), parameter prediction from text (Peters et al., 2025), machine coding and rating scales (Bunt et al., 2025; Eberhardt et al., 2025), and AI-assisted construction and adaptation (Grobelny et al., 2025; Suárez-Álvarez et al., 2026)—promise much psychometric work before any respondent is recruited, yet each inherits its model's run-to-run stochasticity, prompt sensitivity, and black-box opacity (Bunt et al., 2025). Semantic factor analysis complements them all: deterministic and exactly reproducible given a fixed embedding matrix, it emulates no behavior but answers a question simulation does not—how a scale's language is organized.

Critiques and Boundary Conditions

The same capabilities, however, threaten classical psychometrics' response data: the autonomous online "synthetic respondent" of Westwood (2025), a coherent demographic persona with memory of prior answers, passed 99.8% of 6,000 attention-check trials, gave largely internally consistent (correctly reverse-scored) responses to established scales, strategically declined "reverse shibboleth" probes of superhuman capability, shifted aggregate polling on instruction, and—most insidiously—inferred researchers' hypotheses from the materials and biased answers toward them by 22.2 percentage points over the original in a replicated classic survey experiment. Westwood's conclusion—a coherent response can no longer be assumed human—cuts two ways: synthetic coherence is cheap and not validity, no substitute for human evidence; yet semantic factor analysis treats embeddings as measures of item meaning, not simulated respondents, and semantic–behavioral convergence as the empirical question `semanticfa` operationalizes.

Analyzing item language also faces the sharpest critique of language-based assessment: for Uher (2025), psychometrics is pragmatic quantification, not measurement—discrimination and prediction with no traceable connection to the quantities to be measured—its cardinal error mistaking judgments of verbal statements for measurements of the phenomena they describe. Standardized items have no unitary meanings in use—each rater construes a local sample of an item's collective field of meaning—so rating-based nomological networks may substantially reflect the items' *inbuilt semantics*, the relations among their words, rather than the phenomena, and language-model scale design or refinement risks perpetuating the error. The package does not contest the critique: its declared object is the inbuilt semantics—the semantic proposal encoded in the item set—and, with no respondent, it measures no attribute of any person and asserts no attribute-to-score causal path in the sense of Borsboom et al. (2004); it can characterize the proposal and compare it with other evidence, never validate a construct or instrument, but it makes the critique's central conjecture estimable scale by scale: the coherence statistics below quantify how much response structure the wording alone anticipates. High coherence vindicates the critique measurably and grounds pre-data structural screening; low coherence marks behavioral covariance semantics cannot explain—a candidate signature of the divergence conditions above or the language–phenomena breaks Uher (2025) describes. Either way, what items say versus how people answer becomes a reported instrument property, not an unexamined confound; Uher's own constructive program—language technologies on unrestricted verbal self-descriptions—complements rather than competes with this use.

The `semanticfa` Package

Decades before embeddings, Lara et al. (1992) held factor interpretation to be at bottom a semantic problem, proposed predicates and multivalued modal logic to mechanize it, and predicted such automation would become a practical necessity for large problems; language models now supply the empirical geometry of meaning that program lacked. "Semantic factor analysis" was coined twice, independently: by Müller (2026), whose pipeline—static Word2Vec vectors of the 50 IPIP marker adjectives weighted by sample mean responses, reduced by principal components, k-means-clustered—recovered all five

Big Five dimensions at 62.5% average cluster purity in 55 respondents whose confirmatory factor analysis had collapsed; and by us, for the response-free framework presented here (Yanitski & Westbury, 2026). The package generalizes the label in three ways: full item stems and contextual sentence encoders rather than single adjectives; no response data at any stage; and the complete exploratory workflow detailed below rather than cluster-purity assignment.

What separates these demonstrations from a usable methodology is infrastructure. Psycholinguists have long supplied it—witness Mandera et al. (2017)'s precomputed, queryable English and Dutch spaces for colleagues who cannot train their own—and the concern persists: Hussain et al. (2024)'s tutorial holds open, locally runnable models to be a precondition for transparency, reproducibility, and data protection, and shows cosine similarities among personality-item embeddings tracking inter-item and inter-construct response correlations (more closely for larger models) and a low-dimensional projection of construct embeddings largely recovering the Big Five grouping. Psychometric discipline is still missing: Low et al. (2024) note that language-based assessment rarely faces the validity and reliability evaluation demanded of any rating scale, and flag factor analysis on embeddings in place of rating-scale responses as an open direction.

The most complete toolkit to date, the AIGENIE R package (Russell-Lasalandra et al., 2026), implements the AI-GENIE framework for generative psychometrics: a large language model drafts a candidate item pool, the items are embedded, and the EGA and UVA machinery introduced above (plus bootstrap EGA for stability) prunes the pool into a structurally vetted set entirely in silico; a companion function applies the same reduction to user-supplied items.

AIGENIE is generation-centric; its analysis-centric complement, a general-purpose toolkit uniting the methods reviewed above in one reproducible workflow for the exploratory factor analysis of existing scales in semantic space, is the `semanticfa` package (Yanitski & Westbury, 2026): five documented stages in R (R Core Team, 2025)—similarity (keyed, keying-free, and entailment-based encodings, including a signed cross-encoder-scored NLI matrix), retention (embedding-adapted parallel analysis with a multi-criterion consensus), extraction (the expected adequacy and fit diagnostics, plus optional Monte Carlo calibration of every index against matched null embeddings), interpretation (construct anchors, researcher-defined bipolar axes, congruence with theoretical partitions or fitted human solutions, and jingle–jangle screening), and refinement (redundancy flagging, response-free candidate short forms, and vetting of new items against an existing fit)—each detailed with its published basis in the Method section. The pipeline accepts raw item text or a precomputed embedding or similarity matrix; fitted models convert to `psych` objects (Revelle, 2026); and the package ships on CRAN with a worked example for every exported function.

The Present Demonstration

What follows is one fully reproducible study exercising every exported function on the 50-item IPIP Big-Five Factor Markers, descendant of the lexical Big-Five program of Goldberg (1990) and test bed of the work reviewed above (Casella et al., 2024; Jung & Seo, 2025; Milano, Luongo, et al., 2025). After the Method, the Results walk the workflow in analyst order: four embedding-based similarity encodings and a signed NLI matrix from the same 50 items, with the literature's reverse-keying problem treated as a modeling decision whose consequences are measured, not assumed; retention by multi-criterion consensus; a five-factor semantic model with Monte Carlo–calibrated diagnostics; interpretation via anchoring, projection, congruence, and jingle–jangle tools; and refinement via redundancy screening, a response-free short form, and vetting of newly written items. Finally, operationalizing the semantic–behavioral coherence relation proposed above, the semantic solution is compared at factor and item-pair level with a conventional exploratory factor analysis of more than 870,000 respondents to the same items (Open-Source Psychometrics Project, 2018), showing, encoding by encoding, where the two relational structures converge and diverge and what the divergences reveal about the scale's language. Consistent with the convergence-with-divergence evidence above (Feraco & Toffalini, 2025; Kojima et al., 2026), the aim is calibration, not replacement: semantic factor analysis complements response-based analysis, hearing what a scale's language says before, and alongside, what its respondents say.

Method

Materials

The demonstration uses the 50-item International Personality Item Pool (IPIP) Big-Five Factor Markers, a public-domain inventory of ten short, first-person items (e.g., "I am the life of the party.") per domain: Extraversion, Emotional Stability, Agreeableness, Conscientiousness, and Openness/Intellect. The set operationalizes the five-factor structure Goldberg (1990) demonstrated across near-comprehensive pools of English trait adjectives—an ideal test bed for a semantic method, pairing one of psychometrics' most replicated target structures with brief, behaviorally concrete items whose meaning a language model can plausibly capture. The inventory ships in the package as `data(big5)`: item texts, codes (E1–O50), precomputed item embeddings (see the Embeddings subsection), the official ±1 scoring key—under which 18 of the 50 items are reverse-keyed, with Emotional Stability keyed in the Neuroticism direction—and the theoretical domain labels used in all analyses, tables, and figures below, which name the second domain Neuroticism (its keyed direction) and the fifth Openness (its short form). The scoring metadata enter only the explicitly keyed procedures (the `atomic_reversed` encoding and the scoring of the human responses); the primary similarity encoding is keying-free.

The human benchmark is the Open-Source Psychometrics Project's "Big Five Personality Test" raw-data release (Open-Source Psychometrics Project, 2018), an open administration of the same 50 items on a 5-point Likert scale in which 0 codes a skipped item. Of its 1,015,341 response records we removed every record with a missing response or one outside the 1–5 range, retaining $N = 874{,}434$ respondents; the same release has served as a validation corpus in earlier embedding work (Jung & Seo, 2025; Wulff & Mata, 2025). These responses serve *only* as the validation target—a conventional exploratory factor analysis (EFA) of their correlations against which the semantic solution is compared—and never enter the semantic analyses.

Embeddings

The main analyses use precomputed item embeddings from Qwen3-Embedding-8B (Qwen Team, 2025), a 4,096-dimensional text-embedding model. Each of the 50 item texts was embedded once, offline, with last-token pooling, no instruction prefix, and no normalization; the resulting 50×4096 matrix—with texts, codes, domain labels, and scoring key—ships as `data(big5)`, so the analyses need no Python or network access.

Label anchoring and semantic projection also require vectors for non-item words: the five construct names (*extraversion*, *neuroticism*, *agreeableness*, *conscientiousness*, *openness*) and the projection poles (*anxious*, *calm*, *solitary*, *sociable*). A script in the reproduction archive extracted these from a precomputed one-million-word lexicon embedded with the same model, placing them in the items' vector space as the cosine comparisons require.

The package's on-device default for new text is the smaller Qwen3-Embedding-0.6B (Qwen Team, 2025), yielding 50×1024 vectors for the same items, and backs the two demonstrations that must embed text live: the `sfa_embed()` re-embedding check, which compares fresh vectors with same-model offline reference vectors loaded from a NumPy `.npz` archive by `sfa_load_npz()` (the package's import route for external embeddings), and `sfa_item_fit()`'s vetting of candidate items that postdate the archives. Both run through `reticulate` and `sentence-transformers`, which downloads the model from the Hugging Face Hub on first use.

Finally, one similarity matrix is not embedding-based: `sfa_nli_matrix()` scores all item pairs with `cross-encoder/nli-deberta-v3-base`, a natural-language-inference (NLI) cross-encoder trained on SNLI-tradition corpora (Bowman et al., 2015); the next subsection describes its role.

The `semanticfa` Pipeline

The package (Yanitski & Westbury, 2026) exposes five stages as documented R functions (R Core Team, 2025).

Similarity. `sfa_similarity()` turns embeddings into an item-by-item similarity matrix under four encodings: `atomic` (cosines of L2-normalized vectors) and `atomic_reversed` (reverse-keyed embeddings times -1), the pseudo-factor encodings of Guenole et al. (2025), plus two keying-free encodings recovering the negative relations raw cosines suppress (notably for reverse-keyed items)—`squid`, which subtracts the questionnaire-mean embedding (Pellert et al., 2026), and `mean_centered_pearson`, which centers each vector across its dimensions into a genuine correlation matrix (Pokropek, 2026). A fifth, embedding-free route, `sfa_nli_matrix()`, scores pairs with the NLI cross-encoder above (Bowman et al., 2015) as entailment minus contradiction probability, the signed mapping of Hommel and Arslan (2025), and projects indefinite results (signed matrices need not be positive semi-definite), with notice, to a nearby positive-definite correlation matrix (eigenvalues floored, diagonal rescaled to one).

Retention. `sfa_parallel()` adapts parallel analysis (Horn, 1965) to embeddings—null eigenvalues come from similarity matrices of random unit vectors in the embedding dimension, no sample size involved—retaining by default factors above the 95th percentile of 100 null replicates. `sfa_nfactors()` reports the consensus of up to four criteria: parallel analysis; the latent-root-one rule (eigenvalues above one; Horn, 1965); minimum TEFI (Golino, 2026); and EGA (EBICglasso networks with Walktrap detection, via EGAnet; Golino & Christensen, 2026), which best recovers embedding dimensionality (Garrido et al., 2025). `sfa_dimselect()` adapts the embedding-depth optimization of Golino (2026), choosing the nested leading-coordinate depth maximizing $0.70 \times \text{NMI} - 0.30 \times \text{TEFI}_{\text{norm}}$.

Extraction and diagnostics. `sfa()` runs the pipeline end to end from raw text, embeddings, or a similarity matrix, with minimum-residual factoring and oblimin rotation (oblique, in the varimax tradition; Kaiser, 1958) via psych (Revelle, 2026). Fits report Kaiser–Meyer–Olkin (KMO) sampling adequacy, partition-based TEFI (Golino, 2026), root mean square residual (RMSR), common part accounted for (CAF), McDonald's omega per factor (empirical and best-matching theoretical sets), and, given domain labels, a factor-by-construct dominant average absolute loading (DAAL) matrix (Guenole et al., 2025). On the chance-baseline logic above, `calibrate = TRUE` re-derives RMSR, CAF, and TEFI reference distributions from null replicates with no shared semantic structure (Pokropek, 2026); `as_psych()` converts fits to `psych::fa` objects.

Interpretation. Following Wulff and Mata (2025), `sfa_anchor()` tabulates "belonging," a semantic loadings analogue, as cosine similarity to construct anchors (assigned-item centroids or label embeddings), flagging items weakest on their own construct or closer to a rival. `sfa_project()` applies the semantic projection of Grand et al. (2022) introduced above, scores rescaled so that 0 and 1 denote the two poles. `sfa_congruence()` compares a fit with a theoretical partition, human EFA, or correlation matrix on up to five metrics: factorwise Tucker's ϕ (`psych::factor.congruence`; Revelle, 2026); normalized mutual information (NMI) between partitions (Garrido et al., 2025; Golino, 2026); the adjusted Rand index (ARI) (Hubert & Arabie, 1985); a normalized Frobenius matrix similarity; and a disattenuated item-pair correlation (over split-half reliabilities, missing when either is negative). `sfa_jinglejangle()` screens scale sets for jingle–jangle fallacies—label versus item-content similarity per scale pair (Wulff & Mata, 2025, 2026)—flagging by default at $|\text{content} - \text{label}| \geq 0.20$.

Refinement. `sfa_redundancy()` flags locally dependent pairs by faithful UVA (Christensen et al., 2023)—absolute weighted topological overlap on an EBICglasso network of the similarity matrix (Golino & Christensen, 2026), recommended cutoff 0.25—suggesting removals that keep one item per cluster; a direct cosine criterion is a simpler alternative. `sfa_simplify()` builds response-free short forms from the items most representative of each construct's anchor or medoid (Jung & Seo, 2025; Wang et al., 2026), reporting retention and theory-recovery statistics against the original—candidates pending psychometric validation. `sfa_item_fit()` extends the anchoring logic of Wulff and Mata (2025) to vet newly written items against an existing fit: each is embedded with a named model and scored per construct on similarity to the construct name and its items, cross-loading gap to the second-best construct, and near-duplicates above a redundancy cutoff. `sfa_corplot()` draws the similarity matrix as a factor-grouped heatmap; `sfa_itemplot()` maps items to two dimensions by t-SNE (van der Maaten & Hinton, 2008), UMAP, principal components, or classical multidimensional scaling; and `plot()` methods add a scree plot with parallel-analysis overlay and a loadings heatmap.

Analysis Plan

One master script (`reproduce.R`, seed 42) ran the nine demonstration stages; its `R CMD BATCH` transcript supplies every number in the Results.

Similarity construction. The four embedding encodings came from the 50×4096 Qwen3-Embedding-8B matrix, the signed NLI matrix from the item texts; we inspected each encoding's off-diagonal range and, to probe reverse-keying, the similarities among Extraversion item E1, its reverse-keyed sibling E2, and same-keyed E5.

Live-embedding check. The cache was cleared, all 50 items re-embedded with the default Qwen3-Embedding-0.6B model via `sfa_embed()`, and per-item cosines to the `sfa_load_npz()` reference vectors summarized.

Retention. On the mean-centered Pearson matrix we ran parallel analysis (100 iterations, 95th percentile), the four-method `sfa_nfactors()` consensus (parallel, latent-root-one, TEFI, EGA), and the EGA-based depth optimization of `sfa_dimselect()`.

Main fit. `sfa()` fit the primary model on the keying-free `mean_centered_pearson` encoding, chosen before interpretation for its genuine correlation matrix and for matching theory and human data at least as well as every alternative in the encoding comparison. The factor number, first left to parallel analysis, was fixed at five throughout, with minimum-residual extraction, oblimin rotation, and the full diagnostic battery; `calibrate = TRUE` re-estimation (100 Monte Carlo replicates) supplied null reference distributions

(Pokropek, 2026), and `as_psych()` handed the fit to psych.

Interpretation. `sfa_anchor()` ran with both centroid and label anchors, the latter from the 8B construct-name vectors; `sfa_project()` placed all items on two bipolar 8B pole-word axes, stability (*anxious–calm*) and sociability (*solitary–sociable*) (Grand et al., 2022); theoretical-partition congruence was scored by NMI and ARI; and `sfa_jinglejangle()` screened the five subscales (8B item and label vectors, flag threshold 0.20).

Refinement. Redundancy screening used Unique Variable Analysis (threshold 0.25) and, as a check, the direct cosine criterion (threshold 0.85); `sfa_simplify()` built a short form (anchor selection, five items per construct, 50 to 25 items); and `sfa_item_fit()` vetted three new candidate items. Vetting embeds new text, so it ran in the default on-device model's 0.6B space, with the reference model deliberately fit *without* the scoring column (rationale in the Results); the keying-free similarity side is unaffected.

Model-size comparison. The mean-centered Pearson matrix and four-criterion retention comparison were recomputed from same-family 0.6B (1,024-dimensional) and 4B (2,560-dimensional) embeddings of the same items alongside the 8B used throughout.

Visualization. The script regenerates the mean-centered Pearson and NLI similarity heatmaps, item maps under all four projection methods, the scree plot with parallel-analysis overlay, and the loadings heatmap.

Human benchmark. After the Materials cleaning, the 18 reverse-keyed items were rescored ($6 - x$, the standard direction), yielding the keyed 50×50 Pearson matrix; the verbatim (unkeyed) matrix was kept for the keying-free comparisons. The human EFA used `psych::fa` (Revelle, 2026) on the keyed correlations (minimum residual, oblimin, $n = 874{,}434$), factors fixed a priori at the theoretical five (Goldberg, 1990); the Results add a conventional parallel-analysis recommendation for context.

Semantic–behavioral comparison. Coherence was operationalized at three levels. (a) `sfa_congruence()` gave structural congruence of the main fit with the human EFA on all five metrics, the disattenuated one computed separately at the item-pair level and reported as missing rather than suppressed where the keying flip leaves split-half reliability undefined (see the Results). (b) Item-pair agreement was the Pearson correlation of each of the five semantic matrices with the two human matrices over all 1,225 unique item pairs. (c) For the encoding comparison, a five-factor model was fit under every encoding (and the NLI matrix) and scored against theory, the human EFA (mean best-match $|\phi|$ congruence; oblimin factors are sign-indeterminate), the human matrices, and the internal diagnostics, on the metrics defined in the Results.

Open Practices

A self-contained reproduction archive, to be hosted on the Open Science Framework at `https://osf.io/XXXXX` [OSF link to be inserted], regenerates every number, table, and figure in the demonstration from raw inputs. It contains the master script (`reproduce.R`); a full-run console transcript (`reproduce.Rout`) for checking every command against its real output; the precomputed Qwen3-Embedding-8B item and auxiliary embedding archives with the auxiliary-vector generation script; results as an `.rds` object and CSV tables including the encoding comparison, congruence matrix, and both loading matrices; and the regenerated figures. The human response file is not redistributed where licensing forbids it; the script documents the public-release download (Open-Source Psychometrics Project, 2018) and accepts its location via an environment variable. Only two steps require Python or network access—the NLI matrix and the live-embedding demonstrations, which fetch models from the Hugging Face Hub on first run; all factor analyses, retention methods, diagnostics, congruence metrics, refinement analyses, and figures are pure R and run offline.

The reported run used R 4.5.0 (R Core Team, 2025) with semanticfa 0.1.0 (Yanitski & Westbury, 2026), psych 2.6.5 (Revelle, 2026), EGAnet 2.4.1 (Golino & Christensen, 2026), GPArotation 2026.4-1, Rtsne 0.17, uwot 0.2.4, and reticulate 1.43.0, with seed 42. The package is on CRAN, with development sources and issue tracker at `https://github.com/devon7y/semanticfa`; a vignette walks every exported function through a second, independent worked example.

Results

Embeddings and Similarity Encodings

Verbatim transcript excerpts (abridged where noted) trace the analyst workflow, starting with the bundled inventory and its Qwen3-Embedding-8B vectors:

```
> data(big5)   # 50 IPIP Big-Five Factor Markers + bundled Qwen3-Embedding-8B
> str(big5, max.level = 1)
List of 5
 $ items     : chr [1:50] "I am the life of the party." "I don’t talk a lot."
   "I feel comfortable around people." "I keep in the background." ...
 $ codes     : chr [1:50] "E1" "E2" "E3" "E4" ...
 $ factors   : chr [1:50] "Extraversion" "Extraversion" "Extraversion"
   "Extraversion" ...
 $ scoring   : num [1:50] 1 -1 1 -1 1 -1 1 -1 1 -1 ...
 $ embeddings: num [1:50, 1:4096] 0.0361 0.0193 0.0184 0.0269 0.0081 0.0227
   0.0204 0.0234 0.0214 0.0189 ...
  ..- attr(*, "dimnames")=List of 2
> E8 <- big5$embeddings    # 50 x 4096
```

`sfa_similarity()` derives all four Methods-defined encodings from one matrix; `atomic_reversed` sign-flips the 18 reverse-keyed embeddings, and `squid` follows Pellert et al. (2026):

```
> sim_at  <- sfa_similarity(E8, "atomic",
+                            factors = big5$factors, codes = big5$codes)
> sim_ar  <- sfa_similarity(E8, "atomic_reversed", scoring = big5$scoring,
+                            factors = big5$factors, codes = big5$codes)
> sim_sq  <- sfa_similarity(E8, "squid",
+                            factors = big5$factors, codes = big5$codes)
> sim_mcp <- sfa_similarity(E8, "mean_centered_pearson",
+                            factors = big5$factors, codes = big5$codes)
> enc_range
                        min   max
atomic                0.394 0.938
atomic_reversed      -0.777 0.938
squid                -0.267 0.827
mean_centered_pearson  0.394 0.938
```

Off-diagonals confirm raw cosines' narrow positive band (Hommel & Arslan, 2025; Pellert et al., 2026): .394–.938 over all 1,225 pairs, even opposite-keyed ones (Figure 1, left); negatives must be manufactured (sign-flipping, −.777; `squid`, −.267), whereas `mean_centered_pearson` matched the raw-cosine range to three decimals.

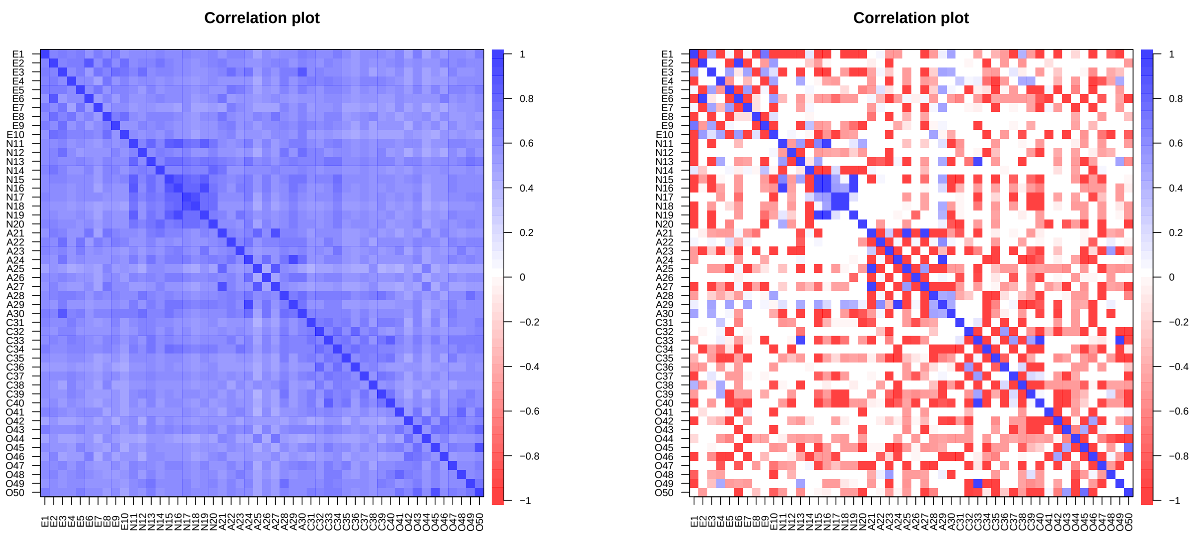

Figure 1

Left: cosine similarity (`atomic` encoding) among the 50 IPIP Big-Five Factor Marker items, grouped by domain (`sfa_corplot()`). All pairwise similarities are positive (range = .394–.938), with darker within-domain blocks (most visibly the Neuroticism items N15–N19) superimposed on a uniformly positive field. Right: signed NLI similarity (entailment minus contradiction) among the 50 items. In contrast to the left panel, contradictions between forward- and reverse-keyed items within a domain produce vivid negative (red) bands and within-block checkerboard structure.

For a signed alternative, `sfa_nli_matrix()` scores ordered pairs with `cross-encoder/nli-deberta-v3-base` in the SNLI tradition (Bowman et al., 2015), symmetrizing entailment minus contradiction (cf. Hommel & Arslan, 2025):

```
> t_nli <- system.time(
+   sim_nli <- sfa_nli_matrix(big5$items)   # cross-encoder/nli-deberta-v3-base
```

```
+ )
NLI matrix: 50 x 50 in 78 s
> round(sim_nli[c("E1", "E2", "E5"), c("E1", "E2", "E5")], 2)
      E1    E2    E5
E1  1.00 -1.00  0.03
E2 -1.00  1.00 -0.98
E5  0.03 -0.98  1.00
> round(sim_ar[c("E1", "E2", "E5"), c("E1", "E2", "E5")], 2)
      E1    E2    E5
E1  1.00 -0.63  0.67
E2 -0.63  1.00 -0.65
E5  0.67 -0.65  1.00
```

NLI scores E1 and its reverse-keyed sibling E2 at −1.00, near-perfect contradiction, versus −.63 under sign-flipping, which reverses vectors without making opposite paraphrases maximally dissimilar (Figure 1, right).

The live-embedding check (Methods) cleared the cache and re-embedded the 50 items with the default Qwen3-Embedding-0.6B:

```
> REF06_NPZ <- "data/Big5_items_0.6B.npz"
> ref06 <- sfa_load_npz(REF06_NPZ)
> ref06
Loaded embeddings (sfa_embeddings)
  Items: 50  |  Dimensions: 1024
  Metadata: codes, items, factors
  Factors: Extraversion, Neuroticism, Agreeableness, Conscientiousness, Openness
  Codes:   E1 E2 E3 E4 E5 E6 E7 E8 ...
> sfa_clear_cache()
> t_emb <- system.time(emb06 <- sfa_embed(big5$items))
sfa_embed: 50 x 1024 in 189.3 s
> print(summary(round(cos_live_ref, 4)))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 0.9999  0.9999  0.9999  0.9999  0.9999  0.9999
```

Every live re-embedding matched its same-model offline reference at cosine ≥ .9999, reproducing the archived output on-device.

Factor Retention

With no respondents to permute, `sfa_parallel()` adapts parallel analysis (Horn, 1965), retaining factors above the 95th percentile of random-embedding null eigenvalues:

```
> pa <- sfa_parallel(sim_mcp, E8)
> pa
Embedding-adapted parallel analysis
  Suggested factors: 5
  Embedding dimension: 4096
  Iterations: 100
  Percentile: 95
```

`sfa_nfactors()` adds three criteria: eigenvalues greater than one (Kaiser, 1960), TEFI (following Golino, 2026), and EGA, validated for embeddings by Garrido et al. (2025):

```
> nf <- sfa_nfactors(sim_mcp, embeddings = E8,
+                    methods = c("parallel", "kaiser", "TEFI", "EGA"))
> nf
Factor retention analysis (embedding-adapted)

  Method      n_factors
  parallel    5
  kaiser      6
  TEFI        2
  EGA         5
  ------------------------
  Consensus   5

Eigenvalues: 29.63   1.92   1.74   1.48   1.27   1.01   0.93   0.88   0.78   0.70  ...
```

Parallel analysis and EGA indicated five factors, Kaiser six, TEFI two; the consensus of five matches the theoretical Big Five, and the dominant first eigenvalue (29.63), the general positivity of embedding similarities, precedes a shallow elbow (Figure 2).

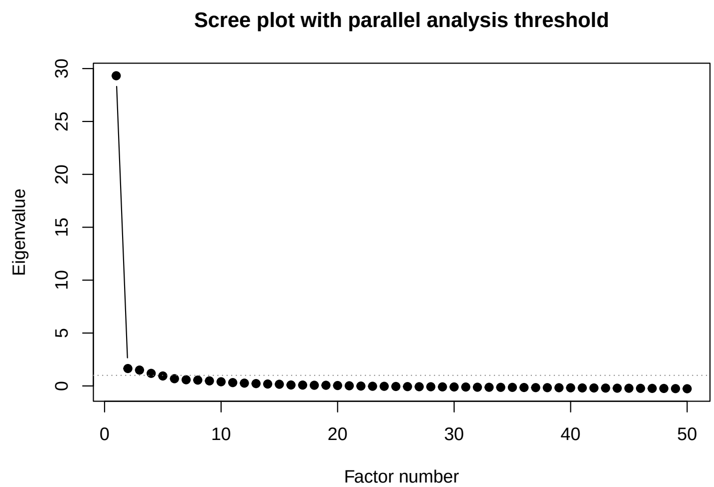

Figure 2

Scree plot of the mean-centered Pearson similarity matrix with the embedding-adapted parallel-analysis threshold (dotted line), produced by `plot(fit, type = "scree")`*. The first eigenvalue (29.63) dwarfs the rest; five eigenvalues clear the parallel-analysis threshold.*

A second question is embedding-specific: how many of the 4,096 coordinates carry structural signal? `sfa_dimselect()` adapts the embedding-depth optimization of Golino (2026), scoring swept truncations by structural recovery (NMI) weighted against entropy fit (TEFI):

```
> ds <- sfa_dimselect(E8, factors = big5$factors,
+                      encoding = "mean_centered_pearson")
> ds
EGA-based embedding-dimension selection
  Method: EGA depth optimization (adapts Golino 2026)
  Full embedding dim: 4096
  Depths evaluated:   148 (range 3-4096)
  Criterion:          0.70*NMI - 0.30*TEFI (normalized)
  Optimal depth:      1431 of 4096 coordinates
    at optimum: n_dim=5  NMI=0.704  TEFI=-23.291
> cat("sfa_dimselect elapsed:", round(t_ds["elapsed"], 1), "s\n")
sfa_dimselect elapsed: 13.6 s
```

The optimum—roughly the first third, 1,431 of 4,096 coordinates—recovered five

EGA dimensions (NMI = .704), the five-factor answer by a different route; the 148-depth sweep took 13.6 s.

Model Size and Structural Clarity

A natural question is how much the embedding model's size matters. To show the gradient directly, the same 50 items were embedded with three sizes of the same model family—Qwen3-Embedding-0.6B (1,024 dimensions), 4B (2,560), and 8B (4,096)—and the mean-centered Pearson similarity matrix was built from each. Figure 3 shows the three matrices side by side, items grouped by domain: the five diagonal blocks sharpen visibly with model size, with the 0.6B matrix showing the haziest block boundaries and the 8B matrix the crispest separation of within-domain from between-domain similarity. Retention sharpens correspondingly. Under the 0.6B and 4B embeddings the criteria disagree—parallel analysis retains four factors, EGA six, for a modal consensus of six—whereas under the 8B embeddings parallel analysis and EGA converge on the theoretical five, and the consensus is five (the eigenvalue-greater-than-one rule reports six and TEFI two at every size). Larger models of the same family thus yield semantic structure that is both visibly cleaner and dimensionally closer to theory, which is why the demonstration's main analyses use the 8B embeddings while the package's on-device default remains the lighter 0.6B model.

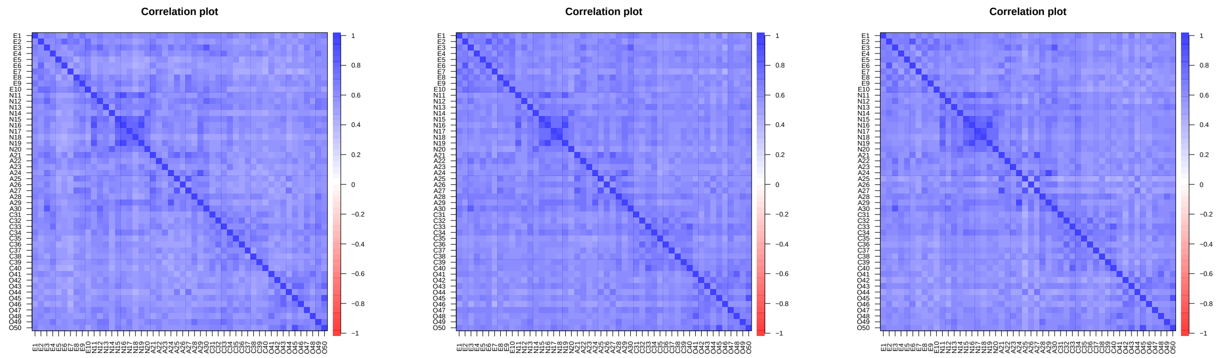

Figure 3

Mean-centered Pearson similarity matrices of the 50 IPIP items, grouped by domain, for Qwen3-Embedding-0.6B (left), 4B (center), and 8B (right). The domain block structure grows progressively crisper with model size.

Model Fit and Diagnostics

`sfa()` runs the whole pipeline in one call; automatic retention chose five factors, fixed as `nfactors = 5`, and the warning confirms the keying-free encoding ignores the scoring column.

```
> fit_auto <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson")
Warning message:
'scoring' has reverse-keyed items but is ignored for encoding =
"mean_centered_pearson": this method is keying-free by design. Use
"atomic_reversed" for keyed sign-flipping.
> cat("Auto-retained factors:", fit_auto$factors, "\n")
Auto-retained factors: 5
> fit <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson",
+            nfactors = 5)
> fit
Semantic Factor Analysis
  Encoding: mean_centered_pearson
  Embedding dim: 4096
  Factors:5 (minres + oblimin)

Diagnostics:
  KMO:  0.971 (marvelous - higher is better)
```

```
  TEFI: -27.7059 (lower is better)
  RMSR: 0.0323 (good - lower is better)
  CAF:  0.4120 (marginal - higher is better)

[item-by-factor loadings omitted; reported in full in the supplement]

                  MR3   MR1   MR2   MR5   MR4
SS loadings     6.421 5.330 4.111 3.193 2.736
Proportion Var 0.128 0.107 0.082 0.064 0.055
Cumulative Var 0.128 0.235 0.317 0.381 0.436

Factor correlations (Phi):
      MR3   MR1   MR2   MR5   MR4
MR3 1.000 0.685 0.606 0.577 0.539
MR1 0.685 1.000 0.538 0.582 0.555
MR2 0.606 0.538 1.000 0.465 0.446
MR5 0.577 0.582 0.465 1.000 0.398
MR4 0.539 0.555 0.446 0.398 1.000
```

KMO was .971, RMSR .032, TEFI −27.71, and CAF .412; substantial factor correlations ($\phi = .40$–$.69$) again reflect the similarities' shared positive component. `summary(fit)` adds the per-factor McDonald's omegas:

```
McDonald's omega per factor:
 factor omega_assigned omega_theoretical matched_theoretical
    MR3          0.908             0.902         Neuroticism
    MR1          0.863             0.850   Conscientiousness
    MR2          0.830             0.560       Agreeableness
    MR5          0.709             0.583        Extraversion
    MR4          0.704             0.732            Openness
```

The solution recovers the Big Five: Neuroticism anchors MR3, Conscientiousness MR1, Openness MR4; MR2 and MR5 split Agreeableness and Extraversion items into warmth/other-orientation and gregariousness factors (Figure 4; Table S1). Communalities ranged from .566 (A23) to .905 (N16); the assigned–theoretical omega gaps—MR2 .830 vs. .560, MR5 .709 vs. .583—quantify this mixing: internally coherent factors that cross the theoretical boundary.

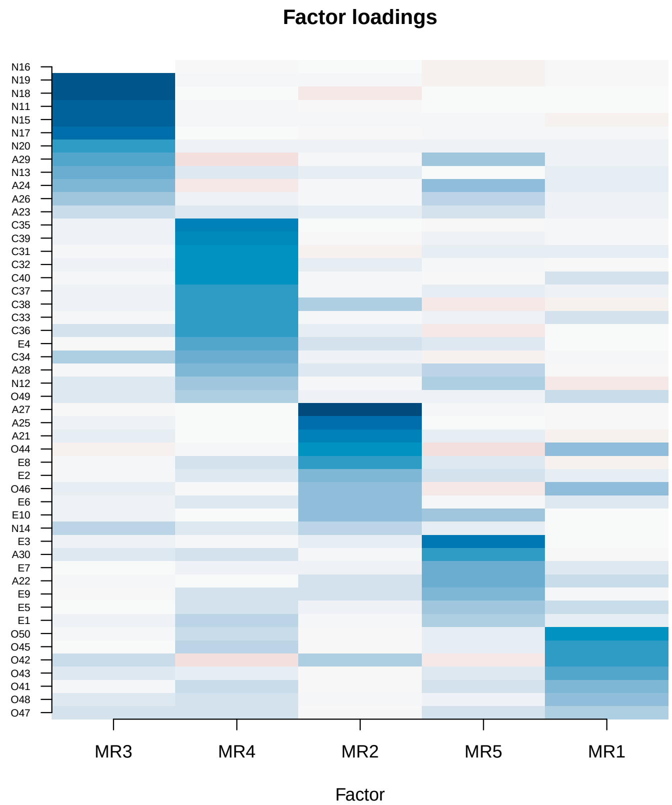

Figure 4

Item-by-factor loading heatmap for the five-factor semantic solution, produced by `plot(fit, type = "loadings")`*. Items are sorted by their dominant factor; MR3 = Neuroticism, MR1 = Conscientiousness, MR2 and MR5 = Agreeableness/Extraversion blends, MR4 = Openness.*

Conventional thresholds need not transfer to embedding-derived matrices (Pokropek, 2026), so `calibrate = TRUE` draws Monte Carlo reference distributions for RMSR, CAF, and TEFI from data-matched null embeddings (100 replicates, 54.7 s):

```
> fit_cal <- sfa(big5_df, embeddings = E8, encoding = "mean_centered_pearson",
+                nfactors = 5, calibrate = TRUE, calibrate_iter = 100)
> cat("calibrate=TRUE elapsed:", round(t_cal["elapsed"], 1), "s\n")
calibrate=TRUE elapsed: 54.7 s
> fit_cal$calibration
[lists of 100 null RMSR, CAF, and TEFI values; omitted]
```

Across the 100 nulls, RMSR ranged from .0119 to .0134, CAF from .501 to .532, and TEFI from −21.66 to −17.82; the observed TEFI (−27.71) undercut every null—a more entropically organized five-factor partition—while the observed RMSR (.032) and CAF (.412) fell above and below their ranges: the real similarities carry residual structure beyond five factors that nulls lack, an index-by-index recalibration that raw thresholds would obscure. `as_psych(fit)` lets `psych::fa.sort()` and `psych::factor.congruence()` treat the fit like a response-based solution.

Interpretation Tools

After Wulff and Mata (2025), `sfa_anchor()` scores each item's cosine to every construct centroid or construct-name embedding, flagging items weak or beaten on their own construct:

```
> anc <- sfa_anchor(fit, anchor = "centroid")
> anc
Semantic anchoring
  Method: Wulff & Mata (2025)
  Anchor: centroid
  Constructs: 5 | Items: 50

[50 x 5 item-by-construct cosine matrix omitted]

  Weakest own-construct similarity (review/removal candidates):
    N12          Neuroticism        own=0.69   <-- higher on Extraversion (0.72)
        "I am relaxed most of the time."
    O44          Openness           own=0.69
        "I am not interested in abstract ideas."
    A23          Agreeableness      own=0.71   <-- higher on Extraversion (0.71)
        "I insult people."
    A25          Agreeableness      own=0.71
        "I am not interested in other people's problems."
    [list abridged]
```

Centroid anchoring mostly flags reverse-keyed items worded nearer another domain—N12 sits nearer the Extraversion centroid (.72) than its own (.69); label anchoring (same Qwen3-Embedding-8B space) flagged an overlapping set, A25 lowest (own .36 vs. .37 on Neuroticism), then E6, A27, C36, N12, and C32.

sfa_project() applies the semantic projection of Grand et al. (2022), placing items on bipolar axes via antonym-pole difference vectors from the same 8B space:

```
> prj <- sfa_project(fit,
+                    axes = list(stability   = c(low = "anxious",  high = "calm"),
+                                sociability = c(low = "solitary", high = "sociable")),
+                    pole_embeddings = poles)
> prj
Semantic projection onto 2 axis/axes
  Method: Grand et al. (2022)
  Scale: 0 = low pole, 1 = high pole

Axis 'stability'  [anxious <-> calm]
  lowest: N13=0.28  N11=0.33  N16=0.37  N19=0.38  N15=0.42
  highest: N12=0.82  A30=0.72  E3=0.65  E4=0.64  C37=0.62

Axis 'sociability'  [solitary <-> sociable]
  lowest: O46=0.37  A27=0.38  A21=0.38  N20=0.39  E4=0.41
  highest: E3=0.82  E7=0.81  A22=0.68  E5=0.68  A30=0.65
```

The projections track content, not scoring key: Neuroticism items fill the anxious pole, yet the most calm-pole item (.82) is again reverse-keyed N12, while party- and conversation-themed E3 and E7, and other-oriented A22 and A30, top the sociable pole.

Versus the theoretical Big Five assignment, sfa_congruence() scored agreement as moderate by NMI and weak by the chance-corrected ARI (Hubert & Arabie, 1985):

```
> cong_theory <- sfa_congruence(fit, target = big5$factors,
+                               metrics = c("nmi", "ari"))
> cong_theory
Factor structure congruence

  NMI:            0.527 (moderate - higher is better)
  ARI:            0.402 (weak - higher is better)
```

sfa_jinglejangle() compares label–label with content–content similarities (Wulff

& Mata, 2025, 2026), here treating the five 10-item subscales as separate scales (8B item and label vectors):

```
> jj <- sfa_jinglejangle(sub_items, item_embeddings = sub_emb,
+                          label_embeddings = jj_lab_emb)
> jj
Jingle-jangle detection across 5 scales
  Method: Wulff & Mata (2025, 2026)
  Flag threshold |content - label| >= 0.20

  No jingle/jangle pairs flagged.
```

Within this single, well-constructed instrument, label and content similarities align; its habitat is cross-instrument screening, where such fallacies are common (Wulff & Mata, 2025).

Response-Free Refinement

Three functions refine a scale before any respondent is recruited. `sfa_redundancy()` detects locally dependent item pairs with Unique Variable Analysis—wTO on an EBICglasso network (Christensen et al., 2023):

```
> red_uva <- sfa_redundancy(fit)                     # UVA / wTO (EGAnet)
> red_uva
Redundant-item detection
  Method: Christensen et al. (2023)
  Measure: wto | threshold: 0.25
  Redundant pairs: 11 | redundant clusters: 10

  Top redundant pairs:
    A24        ~ A29         overlap=0.462
    N17        ~ N18         overlap=0.402
    O45        ~ O50         overlap=0.374
    N16        ~ N19         overlap=0.361
    E2         ~ E6          overlap=0.339
    A25        ~ A27         overlap=0.331
    A21        ~ A27         overlap=0.320
    C32        ~ C36         overlap=0.309
    C33        ~ C40         overlap=0.286
```

```
  O41        ~ O48        overlap=0.268

Suggested removals (keep one per cluster): E6, N19, N18, A25, A27, A29,
  C36, C40, O41, O44, O45
```

Every displayed pair among the eleven (in ten clusters) joins items from one domain; the simpler cosine criterion (threshold 0.85) flagged a largely overlapping set of seven pairs in five clusters, led by N16 ∼ N19 (.938) and A24 ∼ A29 (.933)—near-paraphrases, the "cheating by repeating" that inflates internal consistency without broadening coverage (Christensen et al., 2023).

`sfa_simplify()` builds candidate short forms from the items nearest each construct anchor (Jung & Seo, 2025; Wang et al., 2026):

```
> short <- sfa_simplify(fit, target_n = 5, method = "anchor")   # 50 -> 25 items
> short
Scale simplification (response-free)
  Method: centroid/medoid item selection (in the spirit of Wang et al. 2026;
    Jung & Seo 2025)
  Note: candidate short form -- validate psychometrically before use
  Selection: anchor | groups: theoretical | target_n = 5 per group
  Items: 50 -> 25
  Factors retained (parallel analysis): 5 -> 3
  Structure recovery vs theory (NMI / ARI):
    full:    NMI=0.527  ARI=0.402
    reduced: NMI=0.810  ARI=0.571
  Items kept per construct:
    Extraversion        5
    Neuroticism         5
    Agreeableness       5
    Conscientiousness   5
    Openness            5

  Dropped 25 item(s); see $drop. Reduced fit in $reduced_fit.
> short$keep
 [1] "E2"  "E3"  "E5"  "E8"  "E10" "N11" "N15" "N16" "N17" "N19" "A21" "A22"
[13] "A24" "A28" "A29" "C33" "C34" "C35" "C37" "C40" "O42" "O43" "O45" "O48"
[25] "O50"
```

Halving the scale to 25 items *improved* theoretical-partition recovery (NMI .527 to .810; ARI .402 to .571): pruning semantically peripheral items sharpens boundaries the full pool blurs. Parallel analysis now suggested three factors; per the printed caveat, a semantic short form is a candidate for psychometric validation, not a finished instrument.

`sfa_item_fit()` vets newly written items against an existing fit, extending Wulff and Mata (2025). Live embedding confines the demonstration to the default-model (0.6B) space; the reference fit omits the scoring column for the *anti-topic* reasons given in the Method section.

```
> fit06 <- sfa(big5_df[, c("code", "item", "factor")],
+              embeddings = ref06$embeddings,
+              encoding = "mean_centered_pearson", nfactors = 5)
> cand <- c("I make friends easily.",
+           "I am the life of every party.",
+           "I rarely feel anxious or depressed.")
> ifit <- sfa_item_fit(fit06, cand, model = "Qwen/Qwen3-Embedding-0.6B",
+                      redundancy_cutoff = 0.85)
> ifit
Candidate item fit
  Method: Wulff & Mata (2025), extended

Candidate 1: "I make friends easily."
  [similarity-by-construct table omitted]
  Best construct : Extraversion
  Cross-loading  : gap to 2nd (Neuroticism) = 0.01  ->  ambiguous -
    cross-loads with Neuroticism
  Strength       : about typical for Extraversion (items 0.74 vs avg 0.75)
  Redundancy     : closest item E3 "I feel comfortable around people."
    (0.78) -> distinct
  Verdict        : cross-loads (Extraversion vs Neuroticism) - poor
    discrimination

Candidate 2: "I am the life of every party."
  [similarity-by-construct table omitted]
  Best construct : Extraversion
  Redundancy     : closest item E1 "I am the life of the party." (0.90)
    -> near-duplicate
  Verdict        : redundant (near-duplicate of an existing item)
```

```
Candidate 3: "I rarely feel anxious or depressed."
  [similarity-by-construct table omitted]
  Best construct : Neuroticism
  Strength       : weaker than average for Neuroticism (items 0.75 vs
    avg 0.81)
  Verdict        : weak match - fits Neuroticism worse than its existing
    items (may not belong to this scale)
```

All three planted candidates drew the intended verdicts: cross-loading (Extraversion vs. Neuroticism gap .01), near-duplication of E1 (cosine .90, above the .85 cutoff), and a weaker match to its best construct than that construct's existing items.

Visualization

Beyond the `sfa_corplot()` similarity heatmaps (Figure 1, left and right panels) and the `plot()` scree and loading plots (Figures 2 and 4), `sfa_itemplot()` maps items into two dimensions by t-SNE (van der Maaten & Hinton, 2008), UMAP, PCA, or classical MDS:

```
> for (m in c("tsne", "umap", "pca", "mds")) {
+   sfa_itemplot(fit, method = m, seed = 42, legend = (m == "tsne"))
+ }
```

In all four projections (Figure 5), Neuroticism, Conscientiousness, and Openness items form compact, separated neighborhoods, whereas Extraversion and Agreeableness items interleave in a shared sociability region—a geometric preview of the factor solution's MR2/MR5 blending and the congruence results that follow.

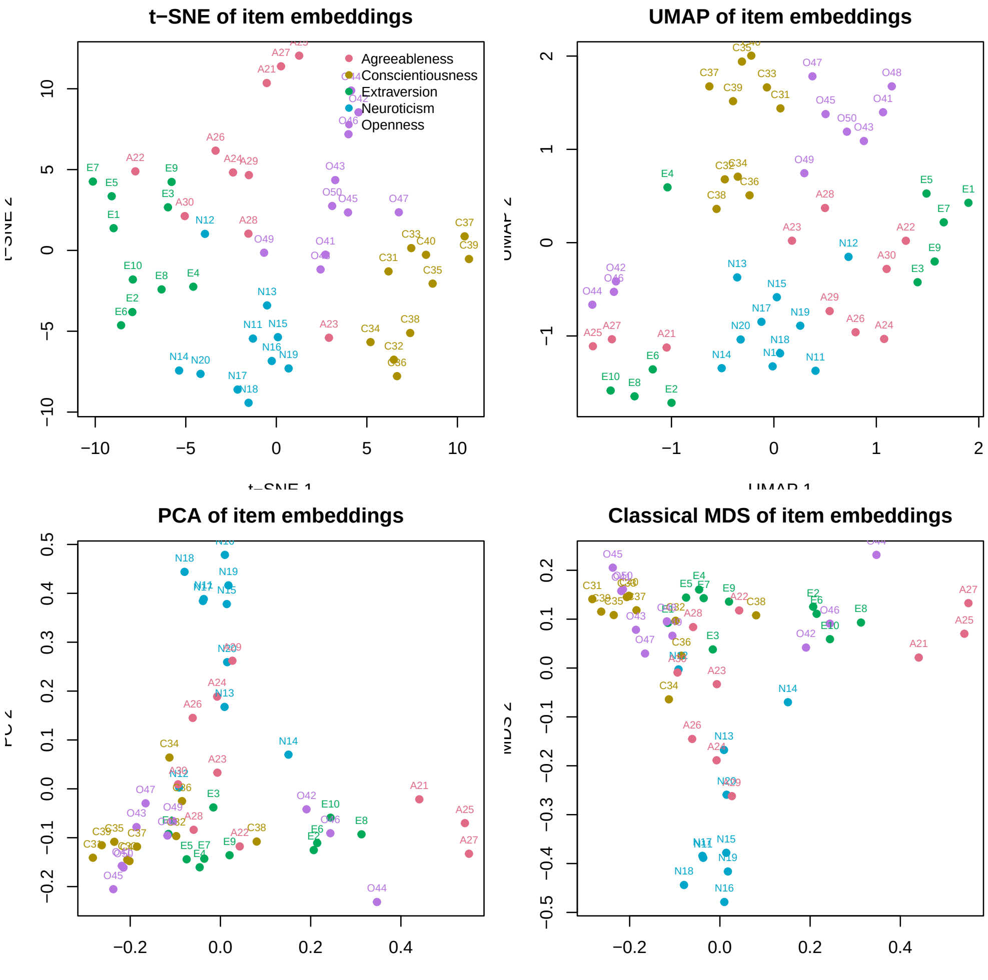

Figure 5

Two-dimensional item maps of the 50 item embeddings produced by `sfa_itemplot()`*: t-SNE (top left), UMAP (top right), PCA (bottom left), and classical MDS (bottom right), with items colored by theoretical domain. Neuroticism, Conscientiousness, and Openness items cluster tightly in all four projections; Extraversion and Agreeableness items interleave.*

Semantic–Behavioral Coherence

Cleaning the human benchmark (see Method) left 874,434 of 1,015,341 respondents, the 18 reverse-keyed items rescored $(6 - x)$ for keyed analyses:

```
> keep <- rowSums(human_raw < 1 | human_raw > 5 | is.na(human_raw)) == 0
> human <- human_raw[keep, ]
> cat("Respondents after cleaning:", n_human, "of", nrow(human_raw), "\n")
```

```
Respondents after cleaning: 874434 of 1015341
> human_fa <- psych::fa(R_human, nfactors = 5, n.obs = n_human,
+                        rotate = "oblimin", fm = "minres")
```

Five factors were fixed a priori (Goldberg, 1990); parallel analysis suggested up to ten—at $N = 874{,}434$ even trivial factors clear noise—but five reproduced the Big Five with simple structure (Table S2).

Factor-Level Congruence

`sfa_congruence()` takes a fitted `psych` model as the target:

```
> cong_human <- sfa_congruence(fit, target = human_fa,
+                              metrics = c("tucker", "nmi", "ari", "frobenius"))
> cong_human
Factor structure congruence

Tucker phi (factor-by-factor):
     MR4   MR1  MR3  MR2  MR5
MR3 0.03  0.80 0.30 0.13 0.12
MR1 0.27  0.11 0.10 0.84 0.28
MR2 0.41 -0.01 0.52 0.05 0.27
MR5 0.64  0.07 0.52 0.11 0.16
MR4 0.18  0.02 0.17 0.14 0.92

  NMI:            0.527 (moderate - higher is better)
  ARI:            0.402 (weak - higher is better)
  Frobenius:      0.664 (moderate - higher is better)
```

Each semantic factor found a distinct human counterpart (Table 1): Openness, Conscientiousness, and Neuroticism strongly ($\phi = .92$, .84, .80); Extraversion (.64) and Agreeableness (.52) weakly, the former matching human Agreeableness nearly as well (.52). Overall, $\bar{\phi} = .744$, NMI = .527, ARI = .402, and normalized Frobenius similarity .664 between the correlation structures.

Table 1

Tucker congruence coefficients between the semantic factors (rows) and the human-response factors (columns)

	Human factor				
Semantic factor	E (MR4)	N (MR1)	A (MR3)	C (MR2)	O (MR5)
N (MR3)	.03	**.80**	.30	.13	.12
C (MR1)	.27	.11	.10	**.84**	.28
A (MR2)	.41	−.01	**.52**	.05	.27
E (MR5)	**.64**	.07	.52	.11	.16
O (MR4)	.18	.02	.17	.14	**.92**

Note. *E = Extraversion; N = Neuroticism; A = Agreeableness; C = Conscientiousness; O = Openness. MR labels are the rotation-order factor names from each solution. Bold values mark each semantic factor's best-matching human factor (mean matched* $\bar{\phi} = .744$*). Semantic solution: minres/oblimin on the mean-centered Pearson similarity matrix; human solution: minres/oblimin on keyed response correlations,* $N = 874{,}434$.

Comparing the Encodings

Table 2 extends this to every encoding's five-factor solution; the NLI matrix, supplied via `similarity =`, was indefinite (minimum eigenvalue −14.1), requiring positive-definite projection (see Method).

Table 2

Encoding comparison: congruence with theory, with the human factor solution, and with human item-pair correlations, plus fit diagnostics

	Theory		Human data				Diagnostics			
Encoding	NMI	ARI	$\bar{\phi}$	r_{keyed}	r_{raw}	r_{dis}	KMO	TEFI	RMSR	CAF
atomic	.527	.402	.744	.434	.456	.707	.971	−27.72	.032	.412
atomic_reversed	.527	.402	.438	.137	.444	—	.971	−27.72	.032	.412
squid	.391	.272	.636	.567	.490	1.000	.022	−44.39	.072	.996
mean_centered_pearson	.527	.402	.744	.434	.456	.707	.971	−27.71	.032	.412
nli	.548	.448	.310	.063	.561	—	.551	−54.15	.094	.688

Note. *NMI = normalized mutual information and ARI = adjusted Rand index of the five-factor item clustering against the theoretical Big Five partition;* $\bar{\phi}$ *= mean Tucker congruence after matching each semantic factor to its best human factor;* r_{keyed} / r_{raw} *= correlation between the encoding's 1,225 item-pair similarities and the human inter-item correlations computed from keyed and raw responses, respectively;* r_{dis} *= disattenuated congruence against the keyed human correlations. Dashes mark cases where the disattenuated metric is undefined and reported as NA because the similarity matrix's split-half reliability is not positive: the keying flip of* `atomic_reversed` *imposes a checkerboard sign pattern (the signed NLI matrix behaves analogously). KMO, TEFI, RMSR, and CAF are the fit diagnostics of each encoding's five-factor solution.*

Three patterns stand out. First, `atomic_reversed`—an anti-topic vector, not reverse-scoring's analogue (see Discussion)—aligned worst among cosines with keyed correlations ($r_{\text{keyed}} = .137$ vs. .434 unflipped) and ordinarily only with raw ($r_{\text{raw}} = .444$).

Second, NLI mirrors the cosines: best on the theoretical partition (NMI = .548, ARI = .448) and on raw pairwise correlations ($r_{\text{raw}} = .561$), worst on loading shape ($\bar{\phi} = .310$), cosine the reverse ($\bar{\phi} = .744$, $r_{\text{raw}} = .456$)—directional entailment versus graded geometry (see Discussion).

Third, SQuID won keyed pair-level agreement ($r_{\text{keyed}} = .567$; r_{dis} saturated at 1.000) yet was essentially unfactorable (KMO = .022), its differencing keeping pairwise relations but stripping extractable common variance. Only keying-free mean-centered Pearson combined near-best theory congruence (within .021 NMI), the best loading-shape match ($\bar{\phi} = .744$), and a genuine, well-conditioned correlation matrix (KMO = .971)—hence the main fit.

Item-Pair Agreement

Across all 1,225 item pairs, structure from language alone matched the raw correlations of 874,434 respondents at $r = .46$ (Figure 6).

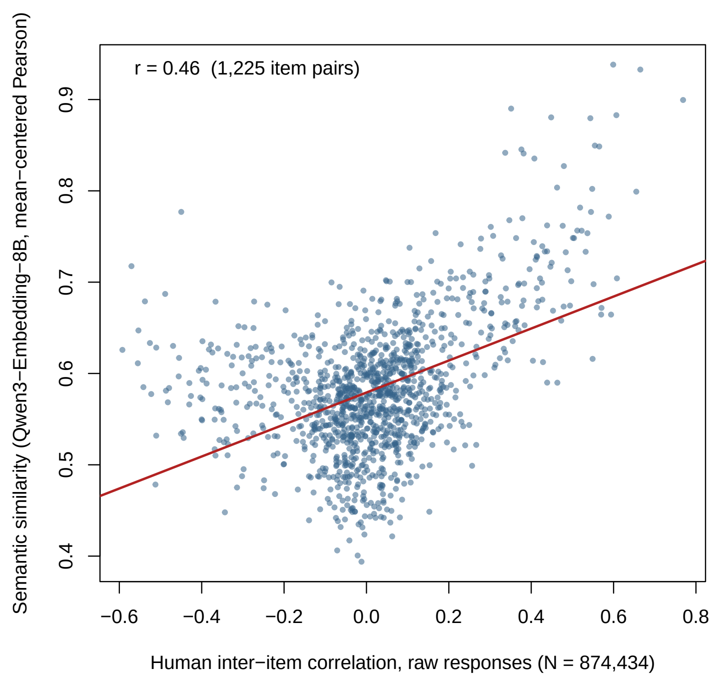

Figure 6

Semantic similarity (mean-centered Pearson encoding of the Qwen3-Embedding-8B item vectors) against human inter-item correlations from raw responses ($N = 874{,}434$), for all 1,225 item pairs; $r = .46$. The fitted regression line is shown in red.

Semantic–behavioral coherence is thus a measured relation: strong convergence for three domains, partial where warmth–sociability language blurs a boundary response covariance keeps sharp, and informative, encoding-specific divergence where the similarity model redefines “meaning the same thing.”

Discussion

Summary of Findings

This article introduced `semanticfa`, an R package that runs the full exploratory factor-analytic workflow—similarity construction, retention, extraction and diagnostics, interpretation, and refinement—on scale items' semantic structure, response-free at every stage. The demonstration ran every exported function on the 50 IPIP Big-Five Factor Markers and validated results against a conventional factor analysis of $N = 874{,}434$ respondents, yielding three findings. First, the instrument's semantic structure is real and largely as intended: a multi-criterion consensus (parallel analysis 5, Kaiser 6, TEFI 2, EGA 5) retained five factors that recovered the Big Five domains, and an independent embedding-depth optimization agreed. Second, semantic–behavioral coherence is graded by domain: matched Tucker congruence with the human solution was .92 (Openness), .84 (Conscientiousness), .80 (Neuroticism), .64 (Extraversion), and .52 (Agreeableness), and item-pair similarities correlated $r = .46$ with raw-response correlations over all 1,225 pairs—useful convergence, informative divergence. Third, encoding item meaning is a substantive modeling decision: sign-flipped reverse-keyed embeddings became anti-topic vectors aligning *worst* with keyed human correlations ($r = .137$ vs. .434 unflipped), entailment- and cosine-based encodings had mirror-image strengths (Table 2), and Monte Carlo calibration showed conventional fit-index cutoffs do not transfer to embedding-derived matrices. Refinement proved most practical: eleven semantically redundant item pairs, a response-free 25-item short form that *exceeded* the full form in recovering the theoretical partition (NMI .527 to .810; ARI .402 to .571), and candidate-item vetting that caught all three planted flaws.

Theoretical Implications

The demonstration shows what estimating the Introduction's semantic–behavioral coherence buys. In the terms of Cronbach and Meehl (1955), the semantic and human analyses are qualitatively different operations—item-meaning geometry without respondents versus 874,434 people's response covariance—so their near-match for Openness, Conscientiousness, and Neuroticism (ϕ = .92, .84, and .80) is intranetwork evidence that both engage the same network region. Under the realist caveat conceded at the outset, this strengthens the evidential webs without certifying that the attributes exist or cause anything (Borsboom et al., 2004): for these domains, the operationalized semantic proposal and response structure are two views of one organization.

The divergences—localized and interpretable, not diffuse—earn the relation its keep: the semantic solution splits Agreeableness and Extraversion into warmth/other-orientation and gregariousness factors rather than losing them, semantic Extraversion thereby matching human Agreeableness (ϕ = .52) nearly as well as semantic Agreeableness did (Table 1)—a seam where English usage runs sociability and warmth together while response covariance holds them apart. The pattern has precedent: Kojima et al. (2026) found near-interchangeability at the general psychopathology factor (congruence = .97) but specific-factor divergences of non-semantic origin (insomnia fused with somatic complaints only in responses, plausibly physiological); Kmetty et al. (2021) found health-related occupations lower in embedding-derived prestige than in survey standing, plausibly via illness connotations; and Zhang and Qiu (2026) found the relation construct-dependent, weakest where baseline inter-item cohesion is strongest. Divergence thus signals its source: semantic compression of distinct behavioral dimensions, connotational drift, or behavioral covariance with causes wording does not carry.

Making coherence estimable also turns the Introduction's divergence conditions into testable predictions. If, as Borsboom (2008) argues, some clinical constructs are causal symptom networks, coherence should run systematically lower there than for common-cause constructs—an ordering `sfa_congruence()` makes falsifiable across instruments of both kinds; the logic extends to formative indicator sets (Bollen & Lennox, 1991) and method variance (Messick, 1995). An unexamined confound of every rating-scale factor analysis becomes a quantity theories can be required to predict.

Methodological Implications

The clearest lesson concerns reverse keying, the Introduction's fault line: a reverse-scored response keeps its content on an inverted scale; an embedding multiplied by -1 is an anti-topic vector. Hence `atomic_reversed` matched keyed human correlations worst of the cosine family (r = .137 vs. .434 unflipped), its checkerboard signs driving split-half reliability negative and rendering disattenuated congruence undefined—a pathology-diagnosing NA, reported, not suppressed. This substantiates the caution that embeddings do not differentiate reverse-keyed items (Suárez-Álvarez et al., 2026), explains why keyed sign-flipping fared worst in HEXACO pseudo-factor analyses (Guenole et al., 2025), fits the positive-sign bias surviving polarity fine-tuning (Hommel & Arslan, 2025), and settles negatively whether an embedding can be meaningfully "reversed" (Pellert et al., 2026). Content-not-key logic recurs package-wide: anchoring and projection both flagged N12 ("I am relaxed most of the time."), reverse-keyed yet semantically the inventory's most calm-pole statement (.82). Three upshots follow: prefer keying-free encodings (mean-centered Pearson, SQuID) or signed NLI over sign-flipping; build candidate-vetting reference fits without the scoring column, keeping centroids purely topical; and drop earlier short-form pipelines' drift-prone manual paraphrasing of reverse-keyed items (Jung & Seo, 2025; Wang et al., 2026), checking substitutions scale by scale.

Second, no single encoding or congruence metric dominates; each operationalizes a distinct inter-item relation—cosine symmetric and graded (Reimers & Gurevych, 2019), entailment directional over ordered premise–hypothesis pairs (Bowman et al., 2015). Profiles mirror accordingly: signed NLI best matched the theoretical partition (NMI = .548) and raw correlations (r = .561) but worst matched loading shape ($\bar{\phi}$ = .310 vs. cosine's .744), while SQuID won keyed pair-level agreement (r = .567) on a near-unfactorable differenced matrix (KMO = .022). What language predicts is thus relation- and encoding-dependent, echoing item-parameter modeling: difficulty is text-predictable (Peters et al., 2025), discrimination largely not (Han et al., 2025), though simulation-based item-characteristic-curve reconstruction recovers it more readily (Ormerod, 2026). Two reporting practices follow: compare like with like—keying-free encodings to raw correlations, keyed to rescored; conflation under- or overstates coherence (NLI: $r_{\text{raw}} = .561$ vs. $r_{\text{keyed}} = .063$)—and report congruence on several metrics (loading shape, partition agreement, pair-level correlation), which can rank encodings oppositely.

Third, semantic-space fit needs matched chance baselines: by survey conventions the RMSR of .032 reads as good yet exceeded the entire structureless-embedding null range (.0119–.0134), the CAF (.412) fell below its nulls (.501–.532), and the TEFI (−27.71) beat all 100 null replicates (−21.66 to −17.82)—two of three conventional readings invert. This generalizes beyond confirmatory modeling the finding that chance meets conventional thresholds alarmingly often in embedding space (Pokropek, 2026) and answers the "no sample size" problem of Suárez-Álvarez et al. (2026): absent an N, Monte Carlo nulls supply empirical reference distributions; we propose such calibrated reporting as standard for semantic factor analyses. Retention repeats the pattern: semantic criteria disagree (parallel 5, Kaiser 6, TEFI 2, EGA 5) as simulations predict (Garrido et al., 2025; Golino, 2026), while human-side parallel analysis at $N = 874{,}434$ endorsed ten factors as trivially small minor factors cleared the noise threshold; such two-sided minor-factor proliferation is precisely the case for consensus-based retention with theory-fixed solutions (Goldberg, 1990).

Applications

The package's natural home is the stage Clark and Watson (2019) place first and weight most heavily—pre-data item-pool refinement—and its clinical analogues: screening large candidate pools, flagging redundancy early, reaching populations and settings where large validation samples are impractical (Kambeitz et al., 2025). The demonstration models that pass: redundancy screening exposed eleven within-domain near-paraphrase pairs (A24 $\sim$ A29 leading, wTO .462)—repetition inflating internal consistency and wording-driven correlations, not content (Christensen et al., 2023; Kojima et al., 2026); vetting correctly typed three new items as cross-loader (gap .01), near-duplicate (cosine .90), and weak match (.75 vs. domain average .81); and response-free shortening halved the instrument yet *improved* theoretical-partition recovery (NMI .527 to .810; ARI .402 to .571)—peripheral, not central, items had blurred construct boundaries (Huang et al., 2025; Kojima et al., 2026). The attenuation paradox (Clark & Watson, 2019) transfers: maximizing semantic homogeneity, like internal consistency, can prune breadth the construct requires, so an anchor-centrality-optimized short form still owes response-based validation—a caveat the package itself prints (Jung & Seo, 2025; Wang et al., 2026).

Second, response data may be scarce, absent, or compromised: Müller (2026) recovered semantic structure at $n = 55$ after the same instrument's confirmatory factor analysis collapsed, and `semanticfa`, needing no responses, lets structural screening precede recruitment. The synthetic respondents of Westwood (2025)—passing 99.8% of attention checks, biasing results toward inferred hypotheses—argue for analysis that touches no response pool and against model-derived structure passing as human evidence; hence the congruence machinery: semantic–behavioral coherence is demonstrated, never presumed—no current language model reproduces the human ability distribution (Liu et al., 2025), and virtual-respondent frameworks disclaim replicating human psychological processes (Lim et al., 2025).

Third, the tools scale across instruments: the jingle–jangle screen, rightly silent on the deliberately well-constructed Big-Five Factor Markers, targets cross-instrument screening, where label–content misalignment is endemic (Wulff & Mata, 2025, 2026), and `semanticfa`'s evaluation-time auditing of existing instruments complements AIGENIE's generation-time vetting of new item pools (Russell-Lasalandra et al., 2026). Finally, the design choices are methodological commitments: an explicit, programmatic embedding pipeline recovers structure that prompted chatbots do not (Huang et al., 2025); open, locally run models make sensitive-item analysis transparent and privacy-respecting (Hussain et al., 2024; Low et al., 2024); pinned model identity freezes the measurement instrument across reanalyses; and the pure-R, offline factor-analytic core (only the NLI matrix and live-embedding demonstrations need Python or the network) runs identically in a data-sovereignty-constrained clinic and a teaching laboratory. We commend the transcript-anchored reporting used here—every number checkable against a captured console transcript—to software papers.

Limitations

The central limitation is definitional: semantic factor analysis characterizes wording, not people; with no respondent, it asserts no causal path and supplies none of the response-process theory Borsboom et al. (2004) make the heart of validity (absent even from response-based factor structure), so coherence must never be reported as a validity coefficient. Of the six construct-validity aspects in Messick (1995), semantic evidence informs content (domain boundaries, representation, redundancy) and partly structural (scoring mirroring the semantic proposal); the substantive, generalizability, external, and consequential aspects require response and outcome data. Even the coherence thesis's strongest support bounds it: in Kojima et al. (2026), response-based factor scores predicted lifetime diagnoses better than text-based ones, textual information being effectively a subset of the behavioral. Hence the calibrated framings: a free, instant pilot study always followed by human data (Hommel & Arslan, 2025); no edge over aggregated expert judgment (Schoenegger et al., 2025); and, in Feraco and Toffalini (2025), no false promises of empirical fit though roughly half of misfit verdicts were false alarms—confidence better grounded than condemnation.

High coherence prices rather than rebuts the critique of Uher (2025), simultaneously grounding pre-data screening and measuring the share of "nomological" structure from ratings already in the items' inbuilt semantics—here $r = .46$ at the pair level, $\bar{\phi} = .744$ for factor shapes. Meaning-as-use also bounds the method: embeddings summarize a community's conventionalized regularities of use (Wittgenstein, 1953, §43), not the local construals raters form, so the idiosyncratic share of response variance is missed by design and perfect coherence is unexpected even from a flawless instrument.

Measurement is also model-dependent: corpus register, training window, and architecture change which human behaviors embedding similarities explain, none optimal across tasks (Mandera et al., 2017); the encoder is thus a declared instrument whose robustness merits reporting. The results rest on one model family (Qwen3-Embedding) and English items; even default full vectors can be suboptimal, signal concentrating in coordinate subsets (Golino, 2026)—witness the optimal depth of 1,431 of 4,096 here. Embedding methods trail human validators on behaviorally but not lexically phrased items (Milano, Ponticorvo, & Marocco, 2025), so behavioral instruments may recover less well than the trait-adjacent Big-Five Factor Markers; psychopathological nuance and hard-to-verbalize inner experience, likely underrepresented in training corpora, leave semantic structure least trustworthy where clinical assessment is hardest (Kambeitz et al., 2025).

Finally, the refinement tools carry caveats. Pruning presupposes reflective indicators—formative indicators are a census, not a sample, and removing one changes the construct (Bollen & Lennox, 1991)—so the measurement model must be settled first. The UVA cutoffs were calibrated on simulated response data, not semantic similarity matrices (Christensen et al., 2023). Semantically close items can still differ in response process (difficulty, extremity, framing)—similarity is not measurement equivalence (Wang et al., 2026)—or in theoretical identity, as with subjective stress versus the cortisol stress response; every purely semantic redundancy or jingle–jangle verdict—this package's included—should, as Wulff and Mata (2026) argue, inform expert deliberation, never replace it.

Future Directions

The most direct extension is comparative: coherence estimated across instruments, domains, and populations until it acquires norms. The demonstration—deliberately well-built, its coherence domain-graded—is one data point; the interesting variation lies between construct types: the symptom-network versus common-cause ordering derived above (Borsboom, 2008), construct-dependence in multilevel analyses (Zhang & Qiu, 2026), and the lexical-versus-behavioral phrasing gradient (Milano, Ponticorvo, & Marocco, 2025) await systematic test with a common toolkit. A second is methodological hardening: encoder robustness reported across embedding models as reliability is across raters (Mandera et al., 2017); embedding-depth selection promoted from diagnostic to tuning step, psychometric signal concentrating in coordinate subsets (Golino, 2026); redundancy cutoffs recalibrated on semantic similarity matrices, not inherited from response-data simulation (Christensen et al., 2023); and Monte Carlo calibration grown into community null baselines indexed by item count, embedding dimension, and encoding, so calibrated reporting (Pokropek, 2026) costs a lookup, not a simulation.

A third extends the pre-data use case across languages: structured generative adaptation performs at or near human benchmarks (Grobelny et al., 2025) but lacks a screen; the package suggests one—source and adapted items embedded in a shared multilingual space, congruence metrics repurposed as a pre-fielding check for structural drift in translation. All three keep the division of labor these results recommend. More than thirty years ago, Lara et al. (1992) predicted that semantic interpretation of factor solutions would have to be mechanized; embedding geometry has made it computable, and `semanticfa` makes it routine. What remains human—on this evidence, necessarily—is what wording cannot carry: response processes, criterion relations, and the judgment that turns a characterized proposal into a validated measure.

Conclusion

Every factor analysis of a rating scale rests on two structures: the covariance of responses and the organization of the language that elicited them. The first has a century of method, the second none until now; `semanticfa` brings the full discipline of exploratory factor analysis—explicit encoding choices, consensus retention, matched-null diagnostics, interpretation, and refinement—to that second structure, respondent-free and exactly reproducible from fixed embeddings. On a well-understood inventory it yielded a five-factor recovery of the Big Five; congruence with the 874,434-respondent structure strong for Openness, Conscientiousness, and Neuroticism, informatively weaker where English merges warmth and sociability; anti-topic vectors, not reverse-scored meanings, from sign-flipped reverse-keyed embeddings; and pre-data refinement that flagged redundant items, sharpened a short form, and vetted candidates. Agreement between the two structures—semantic–behavioral coherence—belongs in the nomological network of any construct measured in language: convergence shows words and respondents organizing the construct alike; divergence, where they do not and why. Neither replaces the other; semantic factor analysis estimates, before any respondent is recruited, how much of any scale's structure is already written in its items.

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Supplementary Materials

Full Loading Matrices

Tables S1 and S2 report the complete oblimin-rotated loading matrices for the five-factor semantic solution (mean-centered Pearson encoding of the Qwen3-Embedding-8B item vectors) and the five-factor human-response solution (minres/oblimin on keyed response correlations, $N = 874{,}434$), as written by the reproduction script to `output/semantic_loadings.csv` and `output/human_loadings.csv`.

Table S1

Semantic factor loadings for the 50 IPIP Big-Five Factor Marker items (mean-centered Pearson encoding, minres extraction, oblimin rotation)

Item	Domain	MR3	MR1	MR2	MR5	MR4
E1	Extraversion	.093	.308	.076	.354	.148
E2	Extraversion	.051	.196	.449	.203	.122
E3	Extraversion	.084	.061	.127	.752	.013
E4	Extraversion	−.001	.546	.206	.161	.008
E5	Extraversion	.037	.205	.117	.385	.257
E6	Extraversion	.093	.175	.432	.065	.190
E7	Extraversion	.005	.110	.120	.509	.182
E8	Extraversion	.062	.220	.570	.169	−.044
E9	Extraversion	−.001	.232	.225	.450	.066
E10	Extraversion	.094	.034	.408	.390	.014
N11	Neuroticism	.817	.073	−.001	.013	.023
N12	Neuroticism	.176	.395	.079	.347	−.093
N13	Neuroticism	.490	.177	.154	.022	.157
N14	Neuroticism	.297	.170	.319	.141	.022
N15	Neuroticism	.801	.074	.069	.054	−.042
N16	Neuroticism	1.003	−.020	.010	−.053	−.029
N17	Neuroticism	.790	.034	−.009	.058	.053
N18	Neuroticism	.860	.027	−.087	.019	.029
N19	Neuroticism	.871	.057	.055	−.071	−.012
N20	Neuroticism	.574	.086	.104	.096	.082
A21	Agreeableness	.149	.039	.718	.128	−.043
A22	Agreeableness	.041	.030	.230	.485	.261
A23	Agreeableness	.267	.193	.146	.211	.113
A24	Agreeableness	.453	−.082	.069	.425	.141
A25	Agreeableness	.084	.000	.786	.025	−.031
A26	Agreeableness	.390	.106	.043	.303	.100
A27	Agreeableness	−.023	.030	.895	.072	−.029
A28	Agreeableness	.077	.459	.171	.308	−.026
A29	Agreeableness	.553	−.158	.073	.390	.119
A30	Agreeableness	.179	.203	.046	.596	−.001
C31	Conscientiousness	.050	.626	−.056	.159	.139
C32	Conscientiousness	.091	.617	.144	.070	−.029

Table S1 (continued)

Item	Domain	MR3	MR1	MR2	MR5	MR4
C33	Conscientiousness	.064	.566	.041	.091	.233
C34	Conscientiousness	.354	.503	.081	−.046	.070
C35	Conscientiousness	.103	.716	.004	−.001	.050
C36	Conscientiousness	.207	.566	.125	−.103	.022
C37	Conscientiousness	.082	.567	.048	.121	.103
C38	Conscientiousness	.085	.567	.336	−.094	−.048
C39	Conscientiousness	.082	.677	−.034	.093	.066
C40	Conscientiousness	.074	.605	.046	−.032	.207
O41	Openness	.063	.263	−.034	.222	.450
O42	Openness	.244	−.122	.350	−.095	.563
O43	Openness	.181	.155	−.027	.169	.554
O44	Openness	−.045	.071	.615	−.128	.403
O45	Openness	.012	.300	−.036	.126	.582
O46	Openness	.155	−.003	.438	−.093	.437
O47	Openness	.220	.205	−.009	.217	.332
O48	Openness	.162	.235	.070	.086	.425
O49	Openness	.164	.351	.100	.098	.250
O50	Openness	.069	.249	−.024	.150	.606

Table S2

Human-response factor loadings for the same 50 items (keyed correlations, $N = 874{,}434$; minres extraction, oblimin rotation)

Item	Domain	MR4	MR1	MR3	MR2	MR5
E1	Extraversion	.712	.042	−.010	−.015	−.012
E2	Extraversion	.713	.086	.053	−.026	.008
E3	Extraversion	.623	−.189	.179	.061	−.068
E4	Extraversion	.755	−.023	−.039	.018	−.027
E5	Extraversion	.714	.012	.123	.067	.034
E6	Extraversion	.522	.005	.073	.010	.235
E7	Extraversion	.721	−.004	.069	.012	−.019
E8	Extraversion	.618	.039	−.109	−.070	.026
E9	Extraversion	.653	.027	−.105	−.050	.103
E10	Extraversion	.684	−.079	−.018	.014	−.019
N11	Neuroticism	−.074	.708	.125	.061	−.055
N12	Neuroticism	−.064	.559	.013	.101	−.004
N13	Neuroticism	−.125	.611	.208	.092	.017
N14	Neuroticism	−.129	.349	.059	−.069	.092
N15	Neuroticism	.005	.505	.004	−.029	−.095
N16	Neuroticism	.027	.746	.031	.005	−.068
N17	Neuroticism	.077	.719	−.021	−.091	.008
N18	Neuroticism	.063	.741	−.035	−.096	.005
N19	Neuroticism	.055	.719	−.175	.041	−.017
N20	Neuroticism	−.219	.574	.017	−.135	.114
A21	Agreeableness	−.049	−.056	.499	−.006	.061
A22	Agreeableness	.282	−.041	.522	−.040	.060
A23	Agreeableness	−.180	−.240	.418	.144	−.071
A24	Agreeableness	−.055	.033	.807	−.007	−.004

Table S2 (continued)

Item	Domain	MR4	MR1	MR3	MR2	MR5
A25	Agreeableness	.060	−.027	.661	−.038	−.002
A26	Agreeableness	−.063	.131	.618	.001	−.074
A27	Agreeableness	.225	−.084	.613	−.035	.015
A28	Agreeableness	.077	−.014	.559	.061	.001
A29	Agreeableness	.019	.100	.700	.019	.033
A30	Agreeableness	.260	−.100	.362	.086	.062
C31	Conscientiousness	.015	.000	−.032	.648	.091
C32	Conscientiousness	−.051	−.035	−.069	.573	−.105
C33	Conscientiousness	−.068	.075	.062	.405	.251
C34	Conscientiousness	.012	−.279	−.010	.556	−.003
C35	Conscientiousness	.073	.009	−.002	.632	−.079
C36	Conscientiousness	−.024	−.088	−.037	.614	−.043
C37	Conscientiousness	−.049	.160	−.004	.581	.040
C38	Conscientiousness	.006	−.152	.097	.465	.049
C39	Conscientiousness	.053	.111	.058	.634	−.054
C40	Conscientiousness	.002	.052	.009	.456	.249
O41	Openness	−.015	−.011	−.058	.035	.596
O42	Openness	−.068	−.189	.014	−.030	.579
O43	Openness	−.014	.107	.091	−.099	.545
O44	Openness	−.095	−.109	.114	−.100	.518
O45	Openness	.166	−.027	−.069	.122	.575
O46	Openness	.011	−.032	.091	−.052	.499
O47	Openness	.019	−.115	−.044	.154	.474
O48	Openness	−.006	.076	−.126	−.024	.564
O49	Openness	−.185	.115	.187	.037	.396
O50	Openness	.122	.017	.008	.000	.662